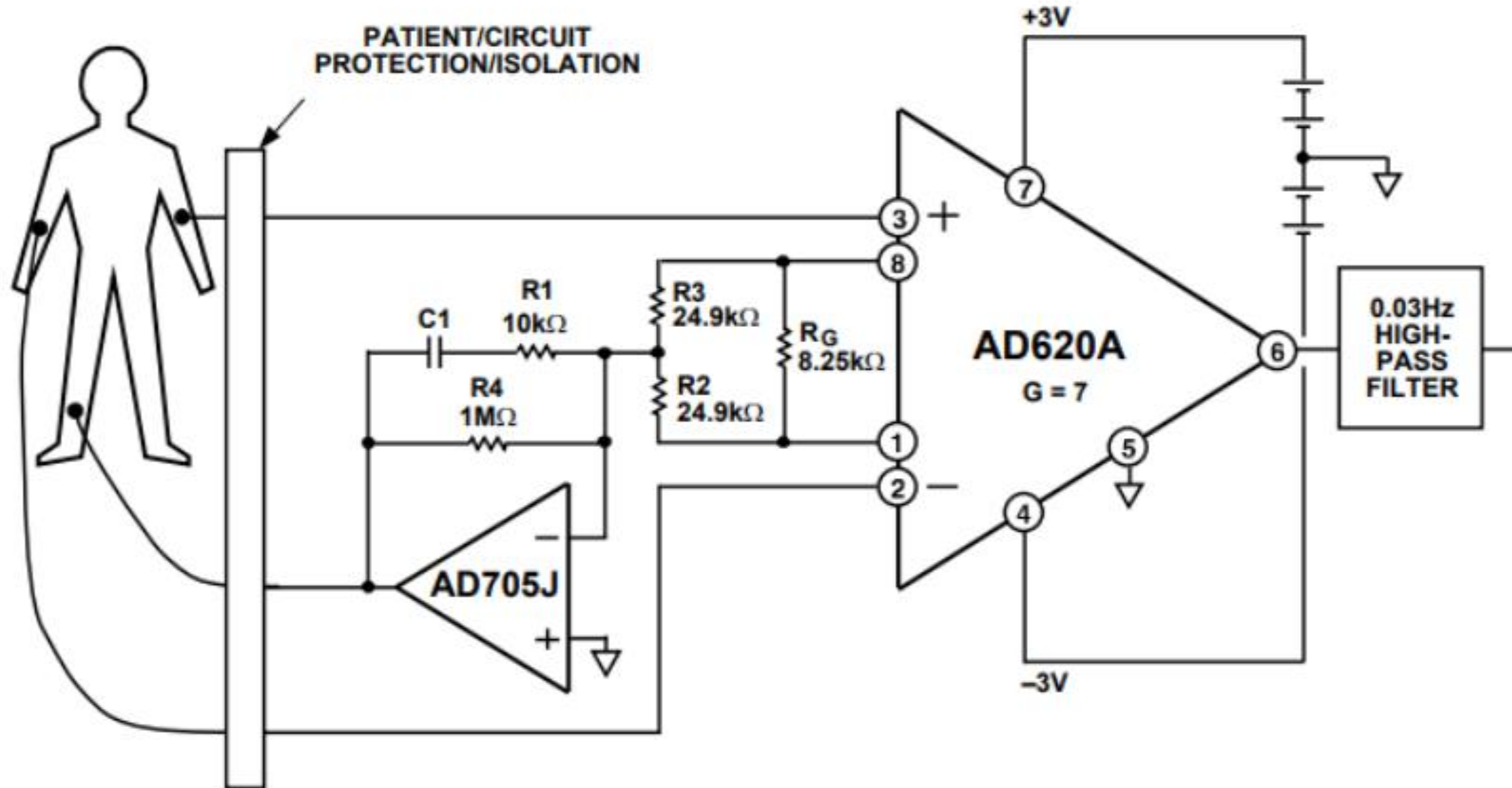


Basic Electronics

Course Overview

- Opamp Basics
- Linear Applications of Opamp and non-idealities
- Non-Linear Applications of Opamp
- First Order Active Filters using Opamp
- ADC and DAC using Opamp
- Number System
- Min Max Terms
- Logic Theorems
- Logic Minimization
- Digital Combinational Circuits



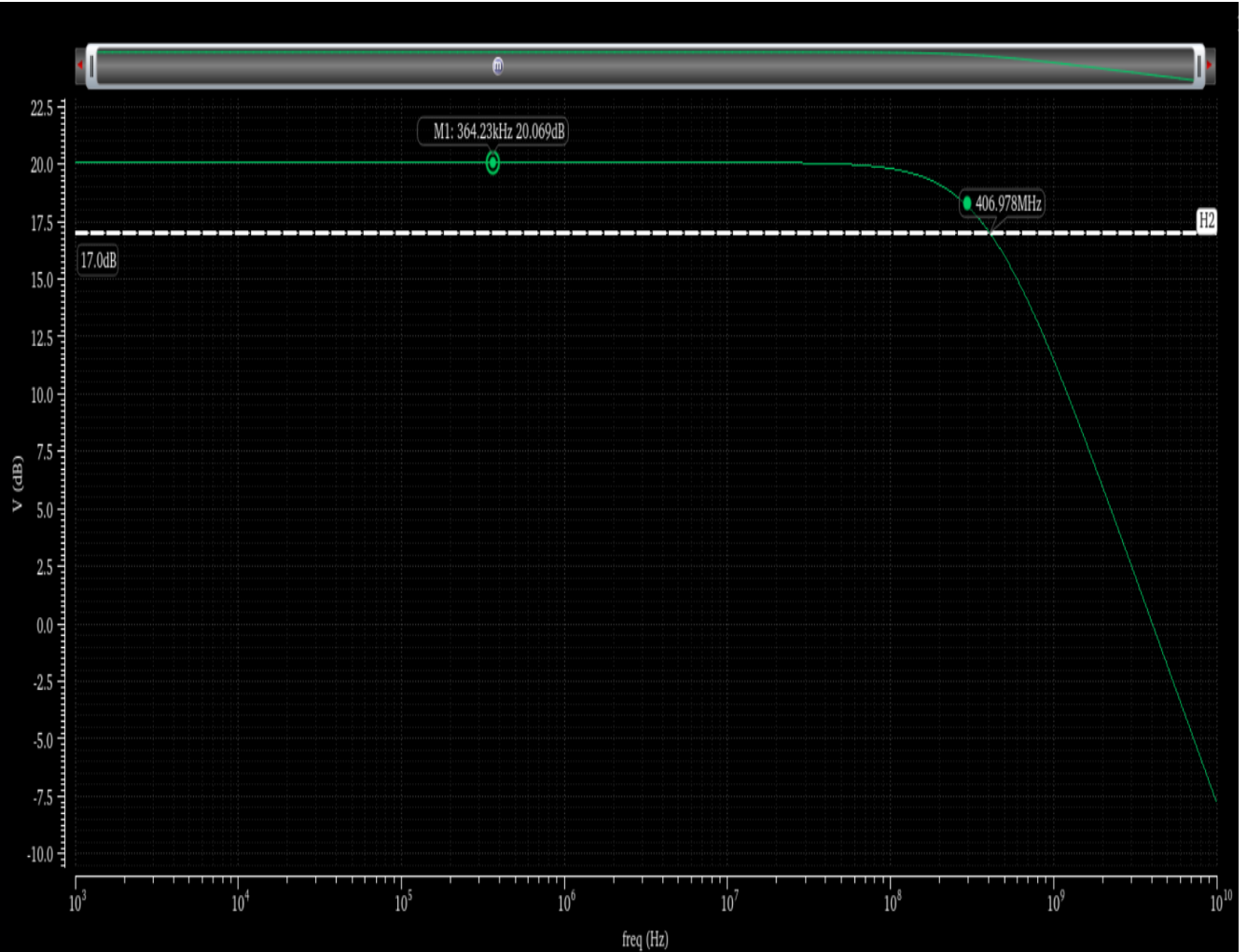
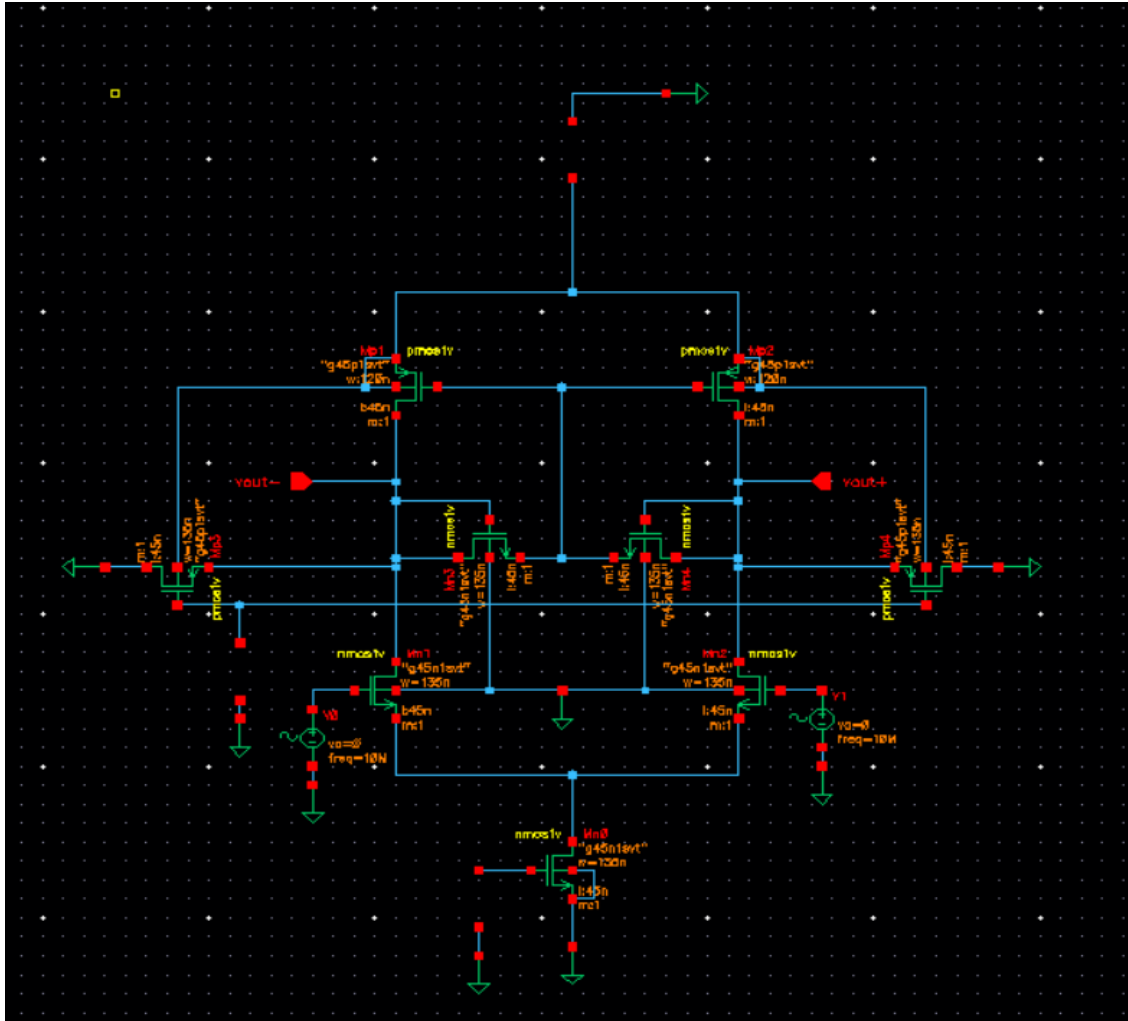
Analog Circuit Design Approaches

Approach – I: quantum mechanics → solid state physics → semiconductor device physics → device modelling → design of circuits

Approach – II: semiconductor device as black box → describe behavior of these black box in terms of terminal I and V → design of circuits

Extremely tedious and time consuming

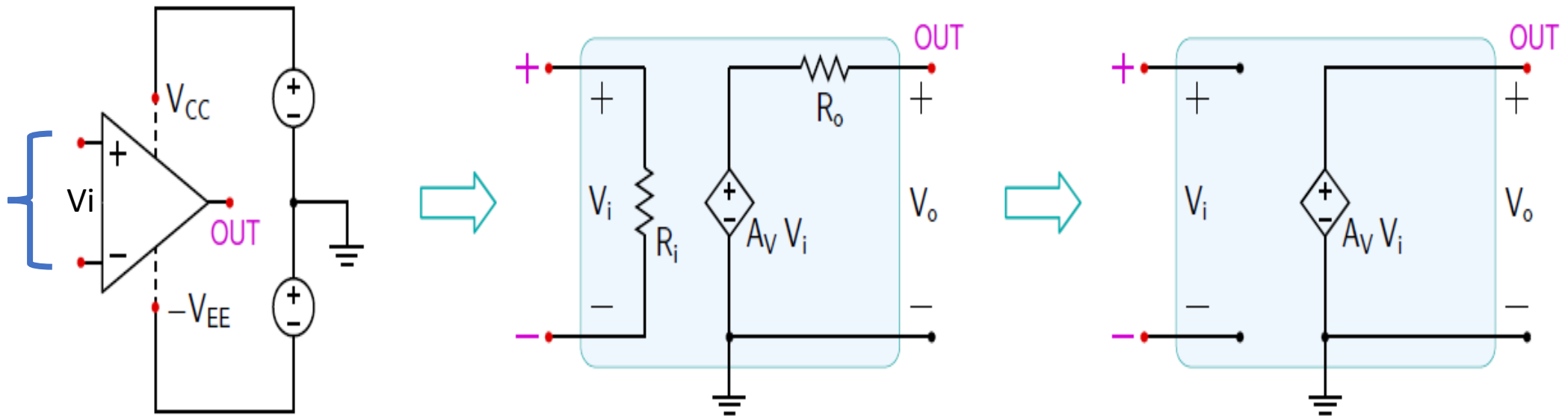
Extremely difficult to identify the cause of the problems

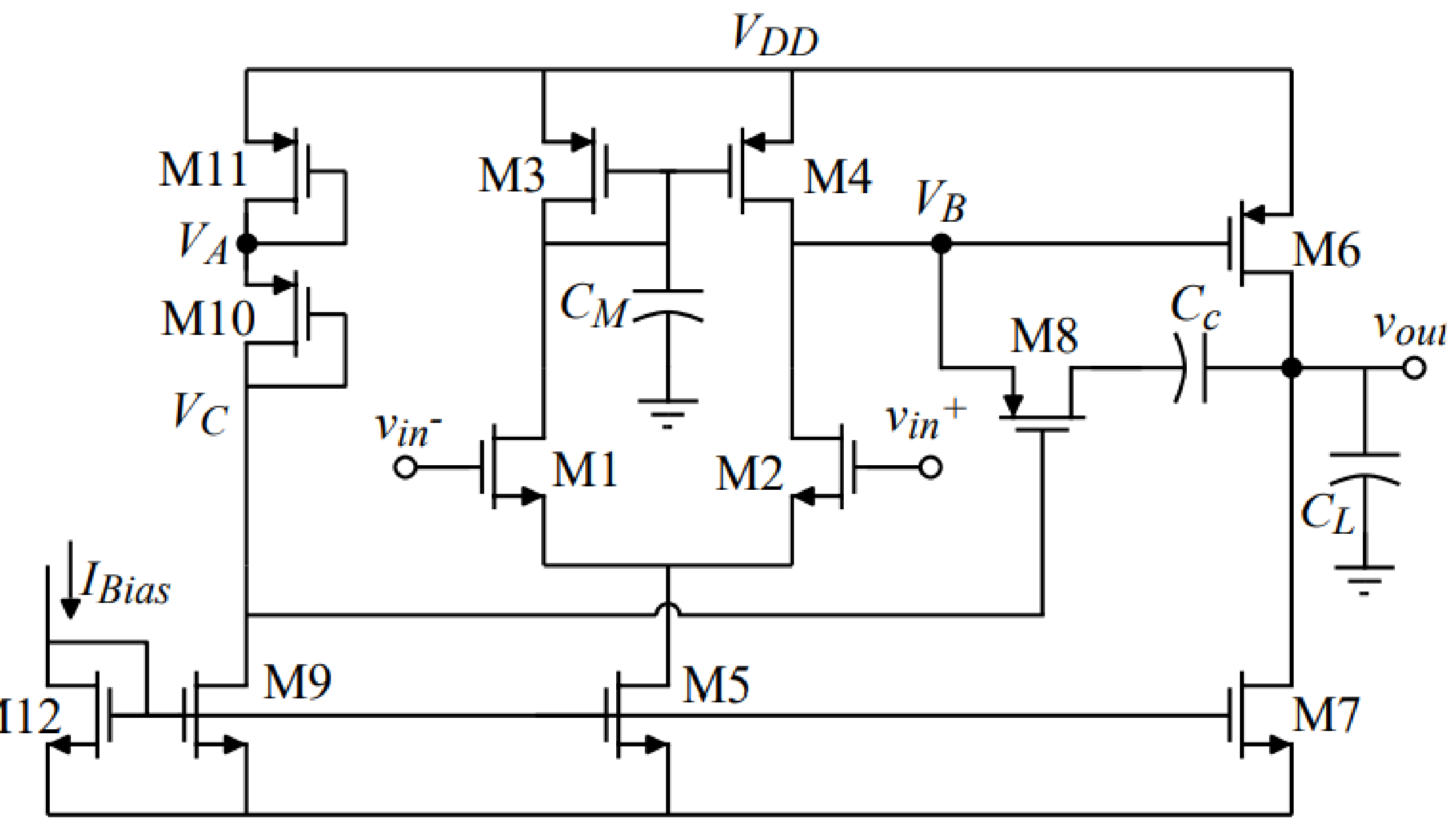


Op-Amp: Equivalent Circuit and Properties

The Operational Amplifier (Op-Amp) is a versatile building block that can be used for realizing several electronic circuits.

- * The characteristics of an op-amp are nearly ideal! Op-amp circuits can be expected to perform according to the theoretical design in most cases.
- * Amplifiers built with op-amps work with DC input voltages as well! useful in sensor applications (e.g., temperature, pressure)
- * The user can generally carry out circuit design without a thorough knowledge of the intricate details of an op-amp. This makes the design process simple.

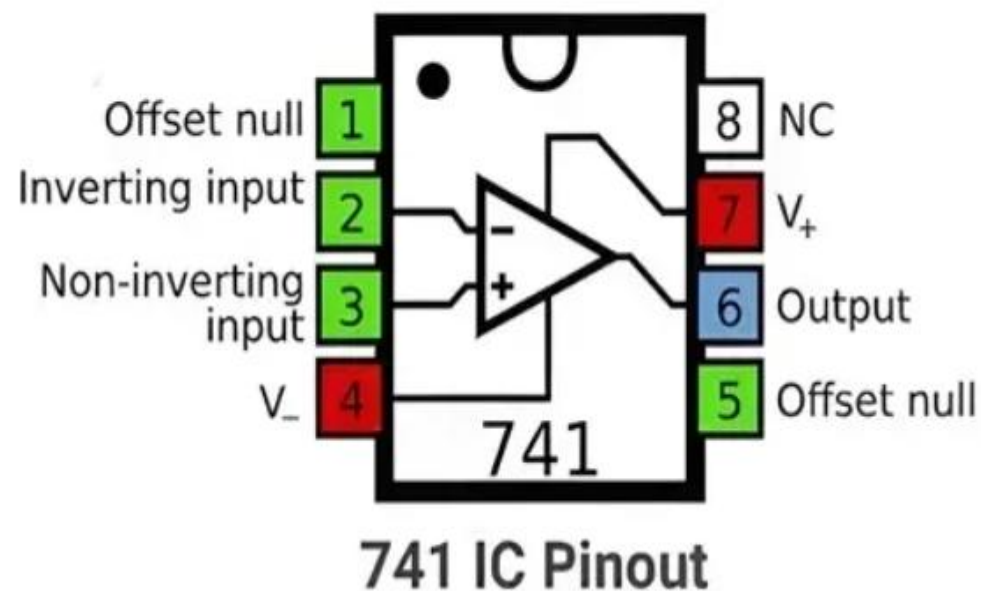




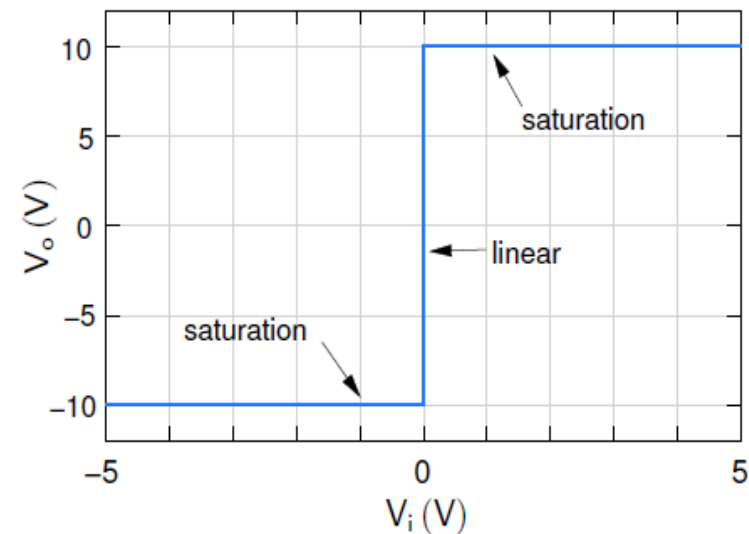
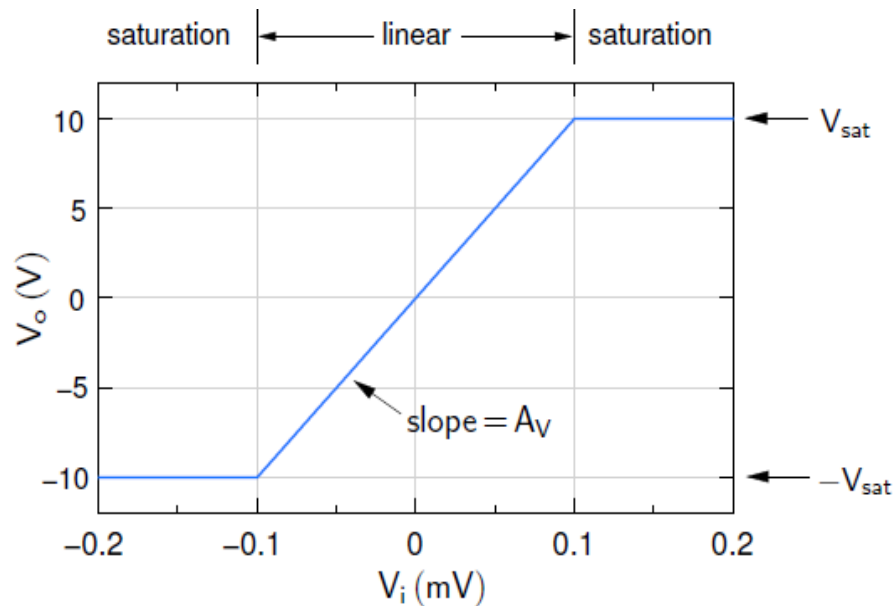
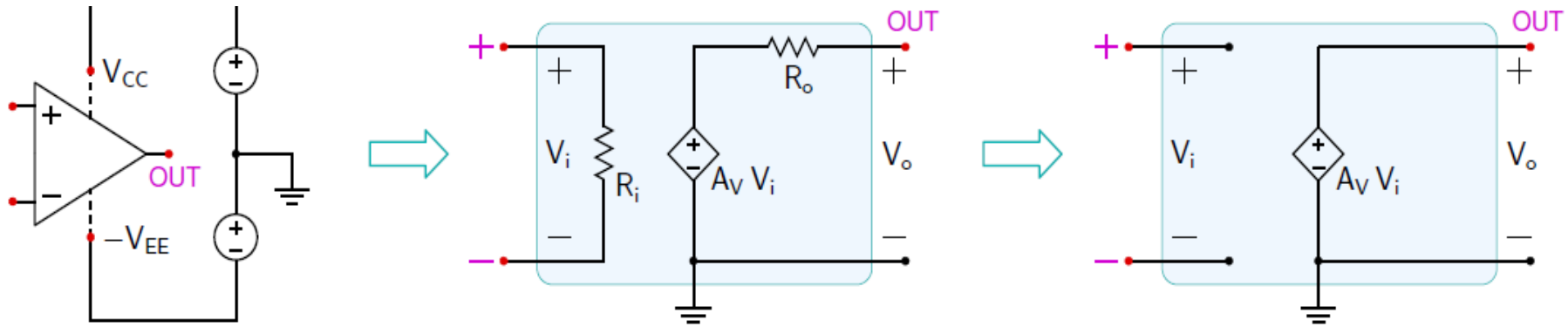
Op-Amp: Equivalent Circuit and Properties

- * The external resistances ($\sim$ a few $k\Omega$) are generally much larger than R_o and much smaller than $R_i \rightarrow$ we can assume $R_i \rightarrow \infty$, $R_o \rightarrow 0$ without significantly affecting the analysis.
- * V_{CC} and $-V_{EE}$ ($\sim \pm 5 V$ to $\pm 15 V$) must be supplied; an op-amp will not work without them!
In op-amp circuits, the supply voltages are often not shown explicitly.

Parameter	Ideal Op-Amp	741
A_V	∞	10^5 (100 dB)
R_i	∞	$2 M\Omega$
R_o	0	75Ω



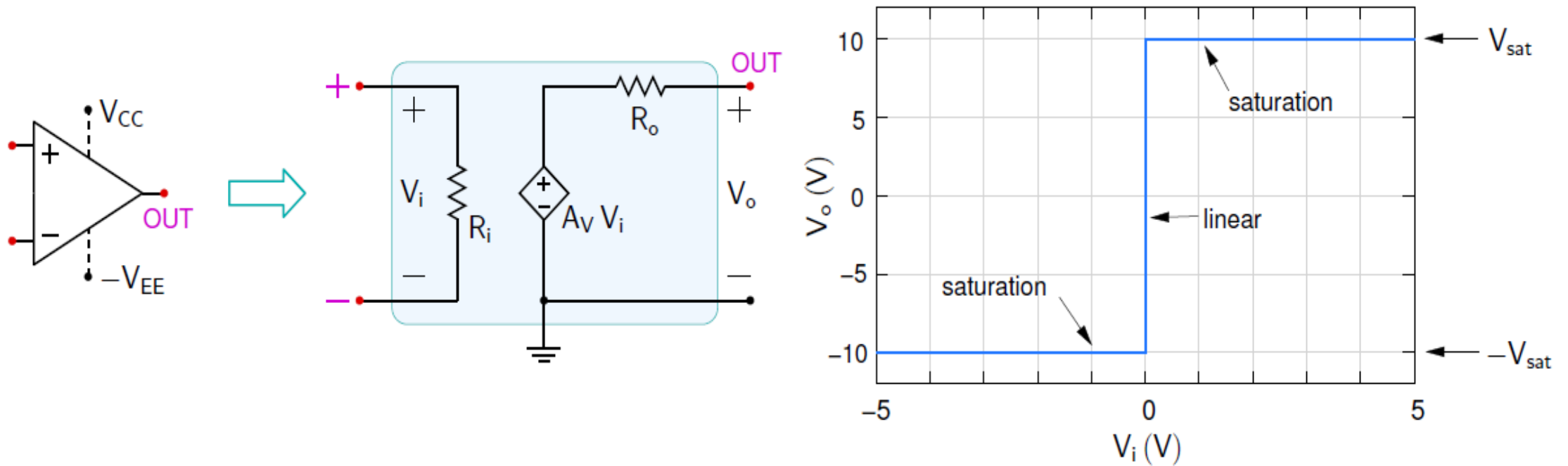
Op-Amp: Equivalent Circuits and Properties



The output voltage V_o is limited to $\pm V_{sat}$, where $V_{sat} \sim 1.5\text{V}$ less than V_{CC} .

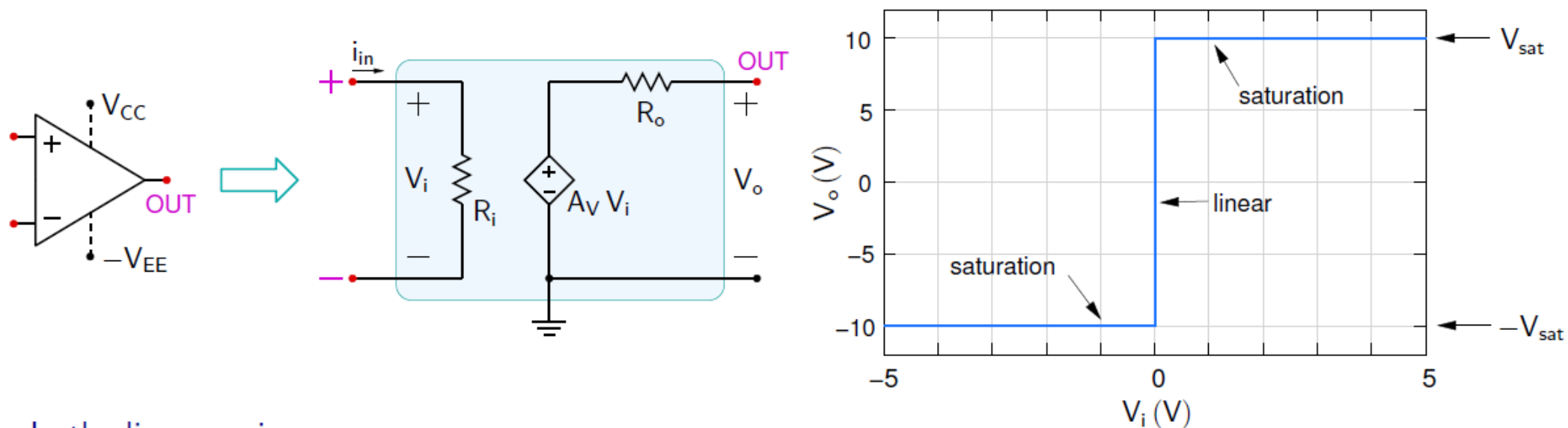
- For $-V_{sat} < V_o < V_{sat}$, $V_i = V_+ - V_- = V_o/A_V$, which is very small.
- V_+ and V_- are virtually the same.

Op-Amp: Equivalent Circuit and Properties



- * Broadly, op-amp circuits can be divided into two categories:
 - op-amp operating in the linear region
 - op-amp operating in the saturation region
- * Whether an op-amp in a given circuit will operate in linear or saturation region depends on:
 - input voltage magnitude
 - type of feedback (negative or positive)

Op-Amp: Equivalent Circuit and Properties



In the linear region,

- * $V_o = A_V (V_+ - V_-)$, i.e., $V_+ - V_- = V_o/A_V$, which is very small

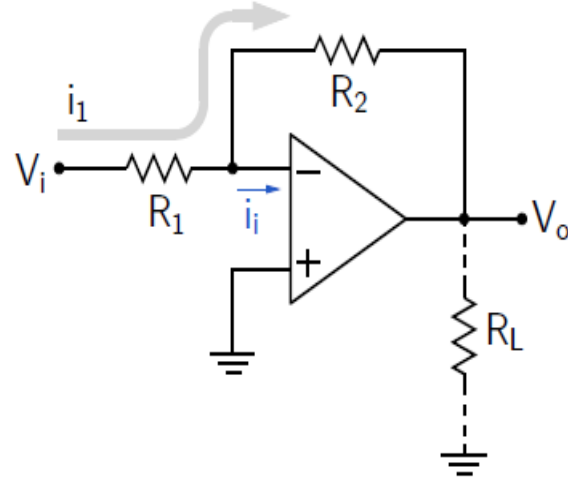
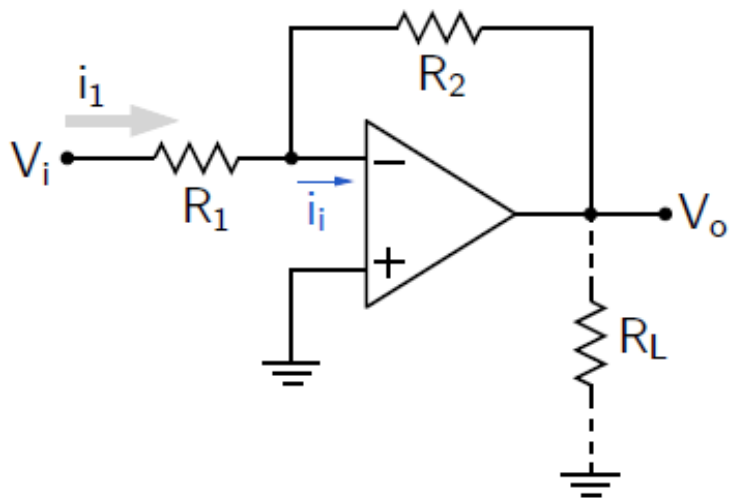
$$\rightarrow \boxed{V_+ \approx V_-}$$

- * Since R_i is typically much larger than other resistances in the circuit, we can assume $R_i \rightarrow \infty$.

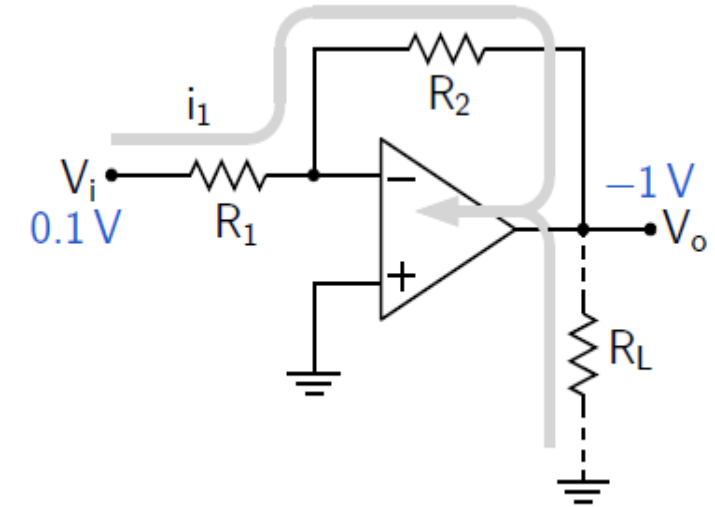
$$\rightarrow \boxed{i_{in} \approx 0}$$

These two “golden rules” enable us to understand several op-amp circuits.

Op-Amp: Linear Application



Where does the current go?



Since $V_+ \approx V_-$, $V_- \approx 0\text{ V}$ so $i_1 = (V_i - 0)/R_1 = V_i/R_1$.

(The non-inverting input is at real ground here, and the inverting input is at virtual ground.)

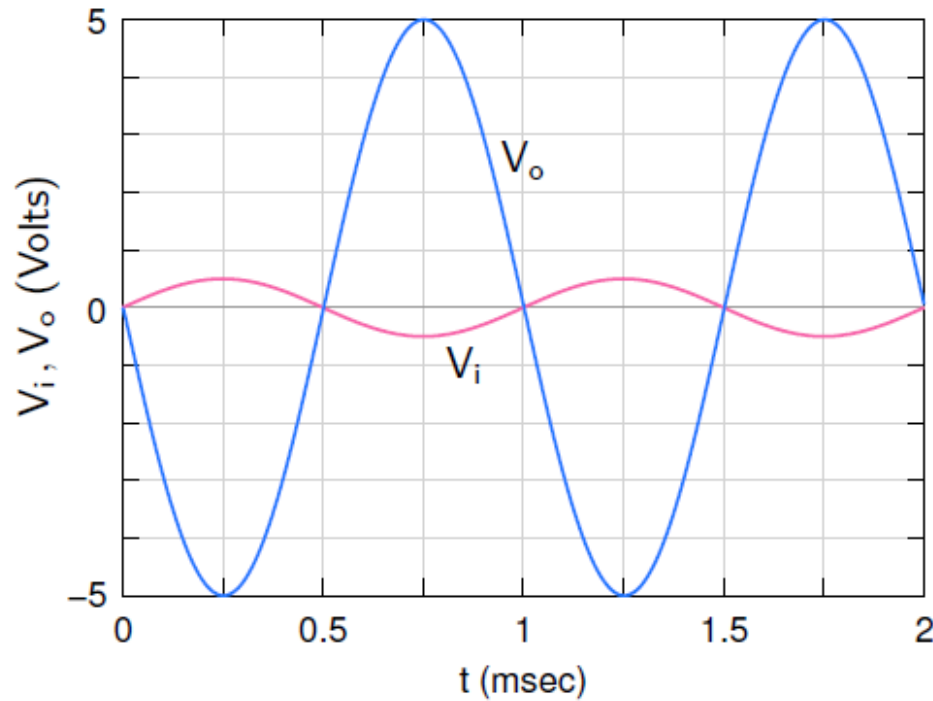
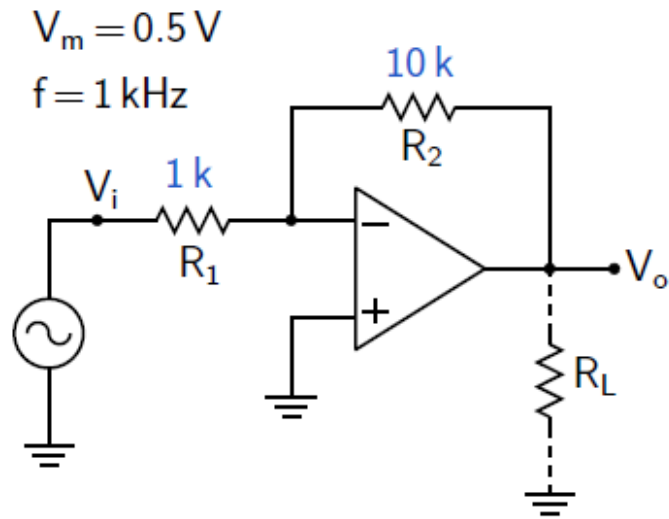
Since I_1 (current entering the op-amp) is zero, i_1 goes through R_2 .

$$\rightarrow V_o = V_- - i_1 R_2 = 0 - \left(\frac{V_i}{R_1} \right) R_2 = - \left(\frac{R_2}{R_1} \right) V_i.$$

"The circuit is called an inverting amplifier"

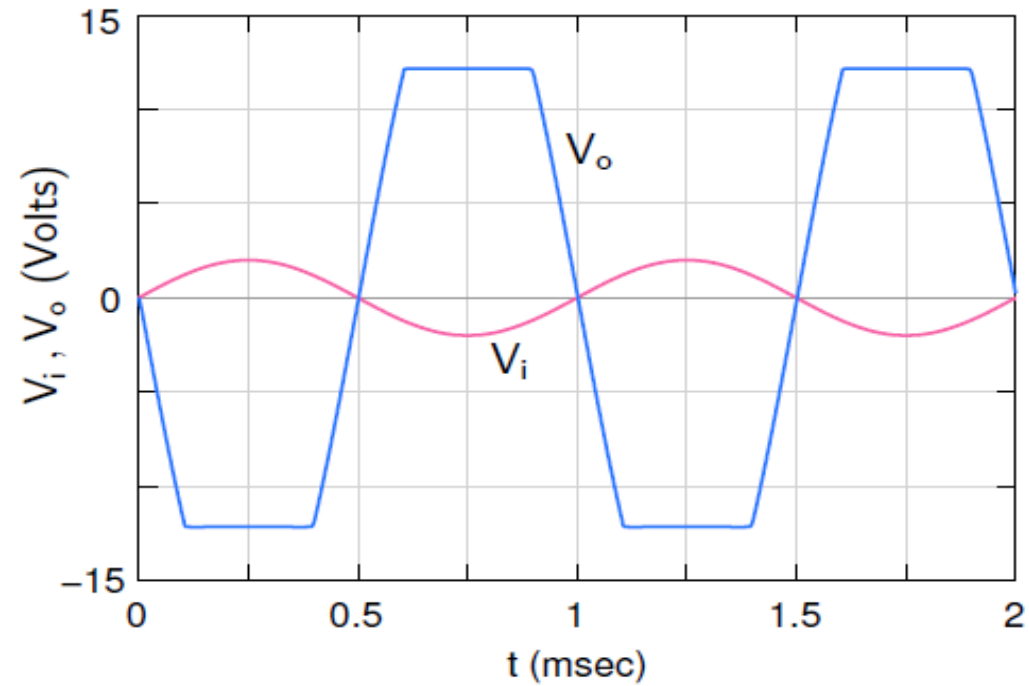
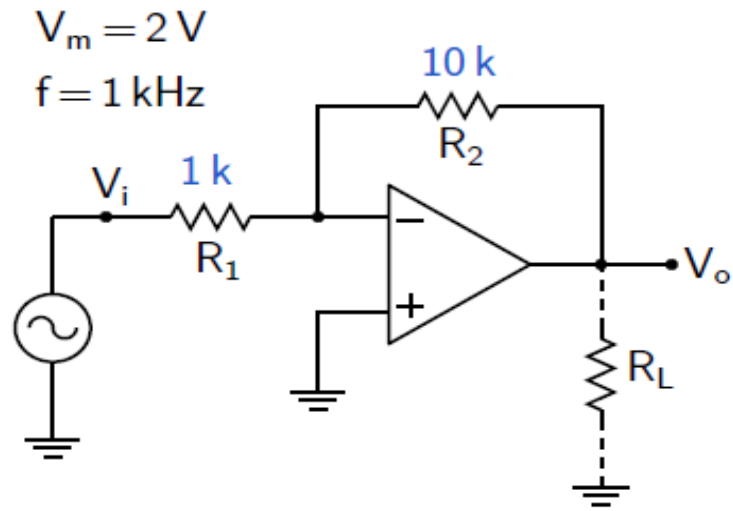
(Op-amp 741 can source or sink about 25 mA.)

Op-Amp: Linear Application



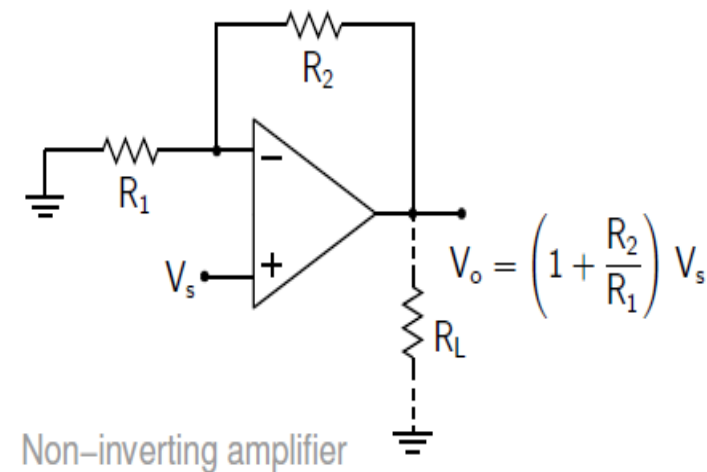
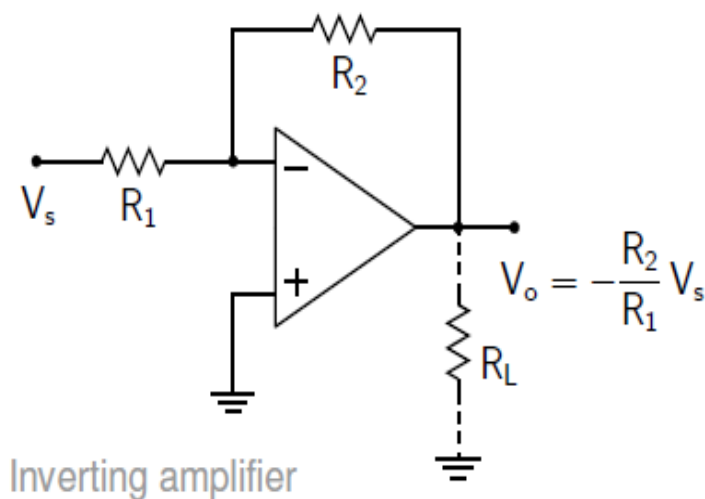
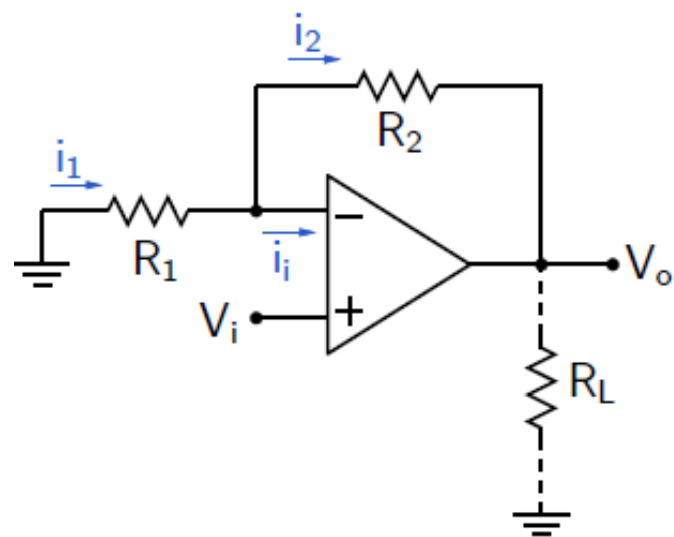
- * The gain of the inverting amplifier is $-R_2/R_1$. It is called the "closed-loop gain" (to distinguish it from the "open-loop gain" of the op-amp which is $\sim 10^5$).
- * The gain can be adjusted simply by changing R_1 or R_2 !
- * For the common-emitter amplifier, on the other hand, the gain $-g_m(R_c \parallel R_L)$ depends on how the BJT is biased (since g_m depends on I_c).

Op-Amp: Linear Application



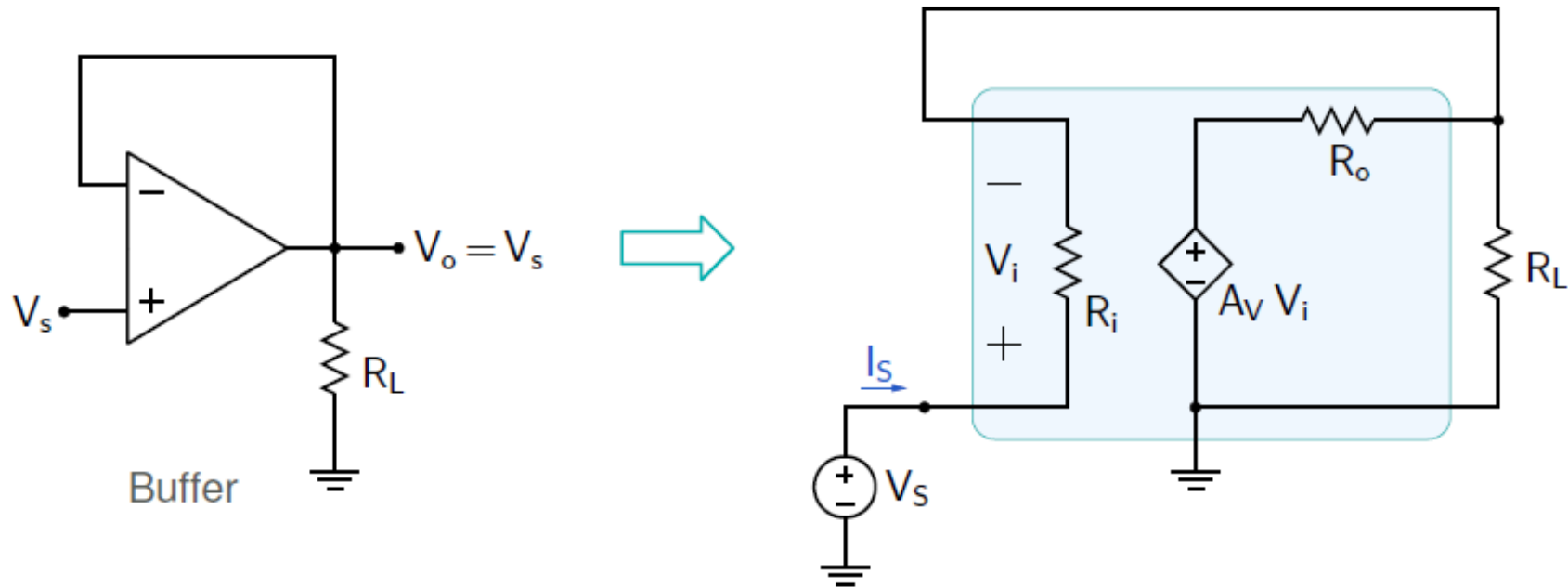
* The output voltage is limited to $\pm V_{\text{sat}}$.

Op-Amp: Linear Application



- * $V_+ \approx V_- = V_i$
 $\rightarrow i_1 = (0 - V_i)/R_1 = -V_i/R_1$.
- * Since $i_i = 0$, $i_2 = i_1 \rightarrow V_o = V_- - i_2 R_2 = V_+ - i_1 R_2 = V_i - \left(-\frac{V_i}{R_1}\right) R_2 = V_i \left(1 + \frac{R_2}{R_1}\right)$.
- * This circuit is known as the “non-inverting amplifier.”
- * Again, interchanging $+$ and $-$ changes the nature of the feedback from negative to positive, and the circuit operation becomes completely different.

Op-Amp: Linear Circuit



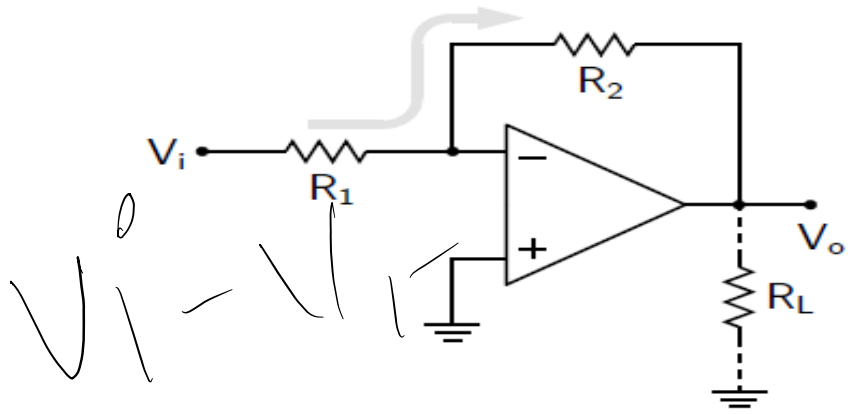
Let $R_o \rightarrow 0$.

$$V_S = V_i + A_V V_i = V_i(1 + A_V).$$

$$I_S = \frac{V_i}{R_i}.$$

$$\rightarrow R_{in} = \frac{V_S}{I_S} = R_i(A_V + 1)$$

Op-Amp: Linear Circuit



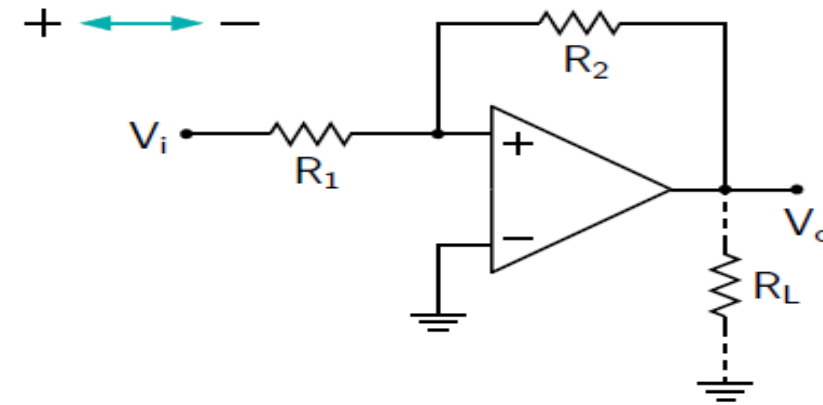
$$V_o = A_V(V_+ - V_-) \quad (1)$$

Since the Op Amp has a high input resistance, $i_{R1} = i_{R2}$, and we get,

$$V_- = V_i \frac{R_2}{R_1 + R_2} + V_o \frac{R_1}{R_1 + R_2} \quad (2)$$

$$V_i \uparrow \xrightarrow{\text{Eq. 2}} V_- \uparrow \xrightarrow{\text{Eq. 1}} V_o \downarrow \xrightarrow{\text{Eq. 2}} V_- \downarrow$$

The circuit reaches a stable equilibrium.

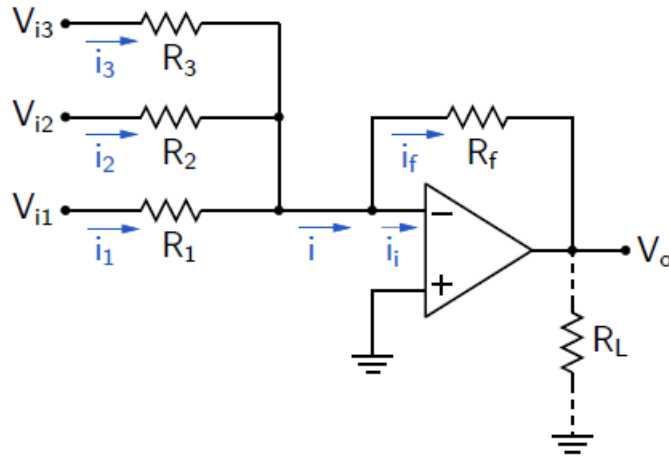


$$V_+ = V_i \frac{R_2}{R_1 + R_2} + V_o \frac{R_1}{R_1 + R_2} \quad (3)$$

$$V_i \uparrow \xrightarrow{\text{Eq. 3}} V_+ \uparrow \xrightarrow{\text{Eq. 1}} V_o \uparrow \xrightarrow{\text{Eq. 3}} V_+ \uparrow$$

We now have a positive feedback situation. As a result, V_o rises (or falls) indefinitely, limited finally by saturation.

Op-Amp: Linear Circuit



$$V_- \approx V_+ = 0 \text{ V} \rightarrow i_1 = V_{i1}/R_1, i_2 = V_{i2}/R_2, i_3 = V_{i3}/R_3.$$

$$i = i_1 + i_2 + i_3 = \left(\frac{V_{i1}}{R_1} + \frac{V_{i2}}{R_2} + \frac{V_{i3}}{R_3} \right).$$

Because of the large input resistance of the op-amp, $i_i \approx 0 \rightarrow i_f = i$, which gives

$$V_o = V_- - i_f R_f = 0 - \left(\frac{V_{i1}}{R_1} + \frac{V_{i2}}{R_2} + \frac{V_{i3}}{R_3} \right) R_f = - \left(\frac{R_f}{R_1} V_{i1} + \frac{R_f}{R_2} V_{i2} + \frac{R_f}{R_3} V_{i3} \right),$$

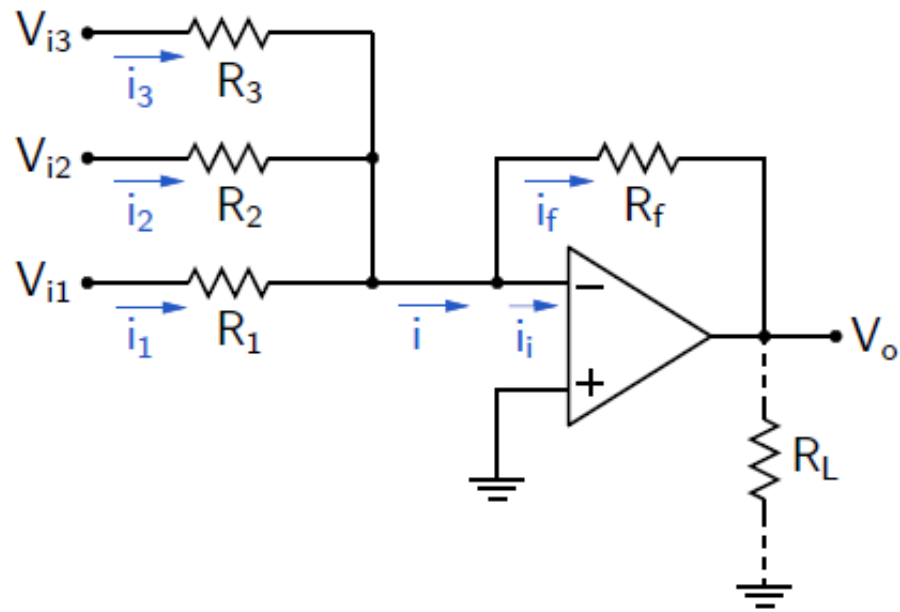
i.e., V_o is a *weighted sum* of V_{i1} , V_{i2} , V_{i3} .

If $R_1 = R_2 = R_3 = R$, the circuit acts as a summer, giving

$V_o = -K (V_{i1} + V_{i2} + V_{i3})$

 with $K = R_f/R$.

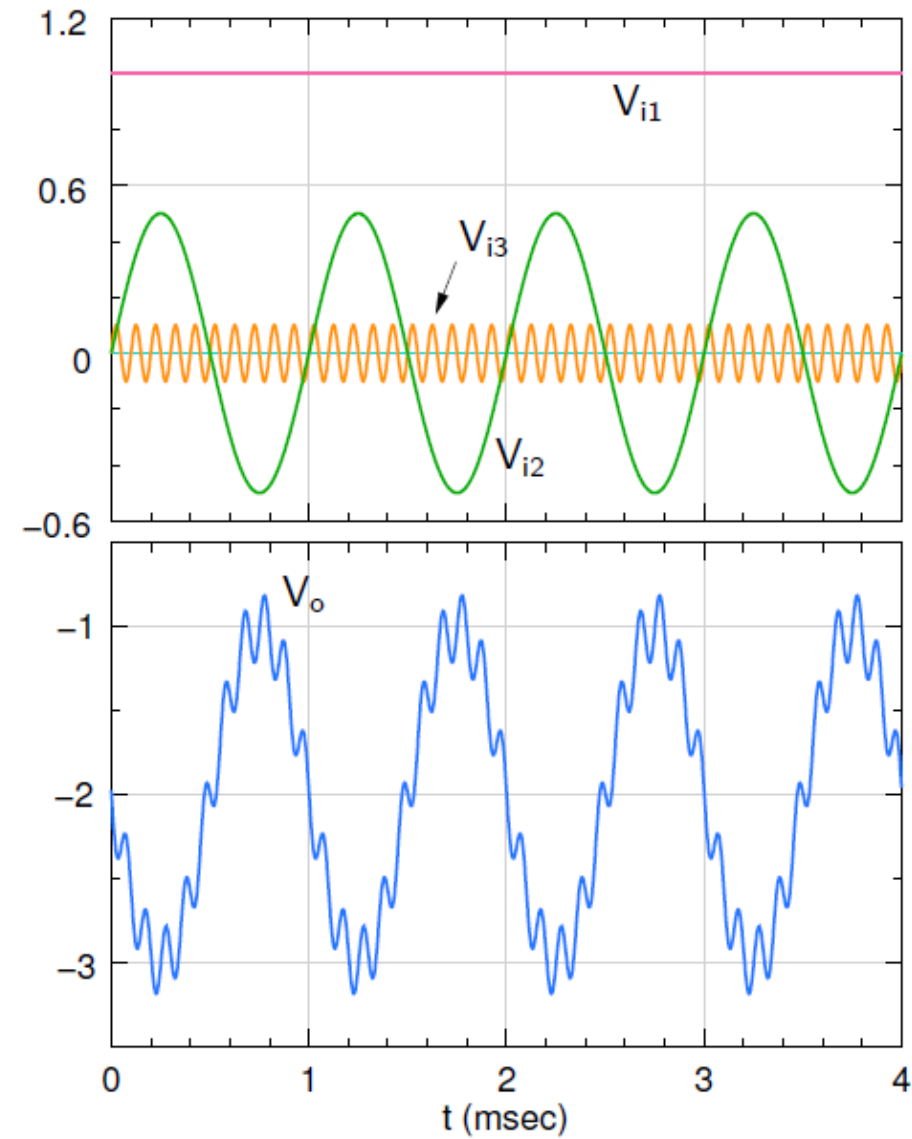
Op-Amp: Linear Circuit



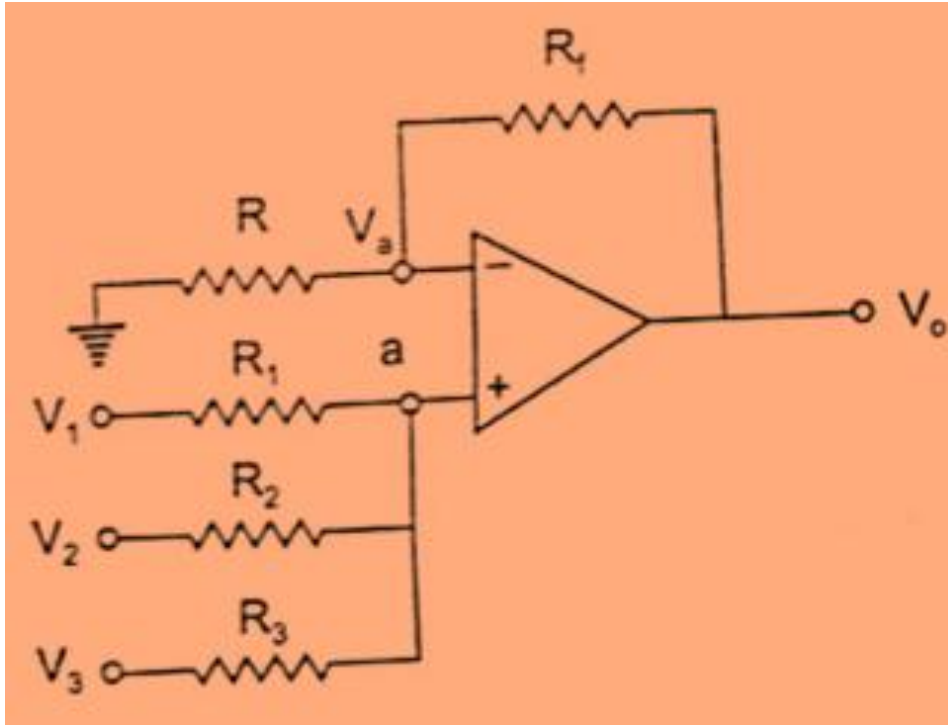
$$R_1 = R_2 = R_3 = 1 \text{ k}\Omega$$

$$R_f = 2 \text{ k}\Omega$$

$$\rightarrow V_o = -2 (V_{i1} + V_{i2} + V_{i3})$$

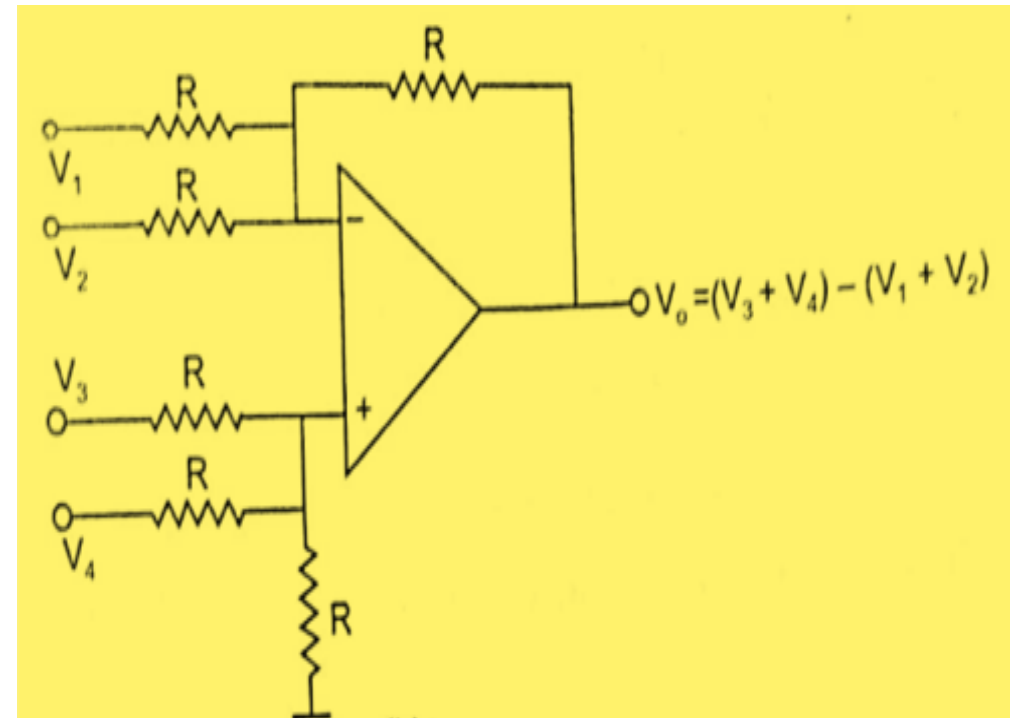
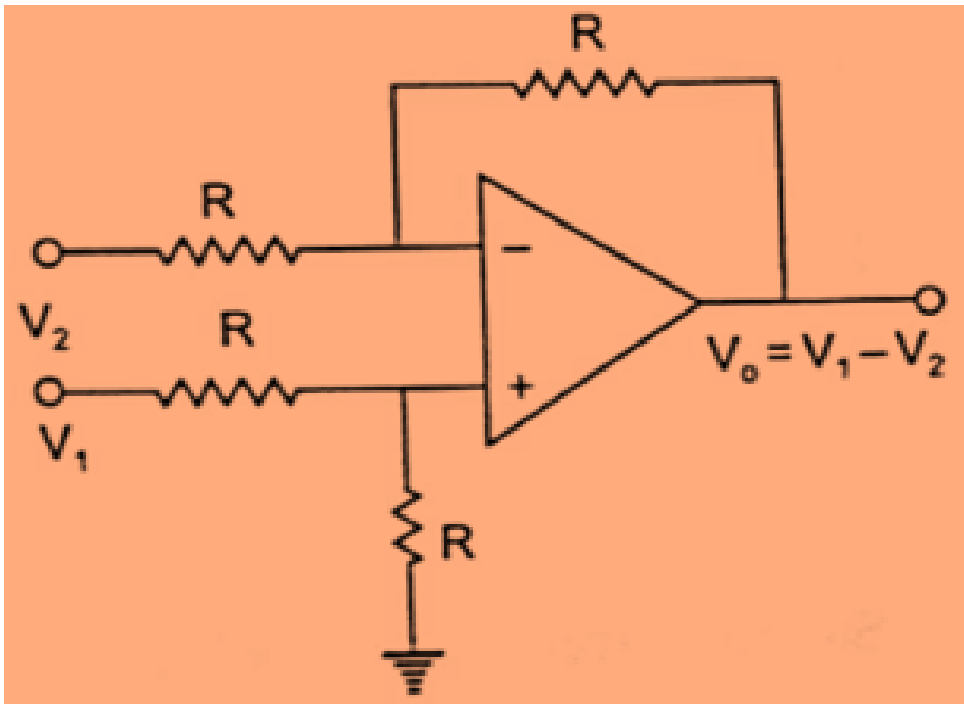


Op-Amp: Linear Circuit

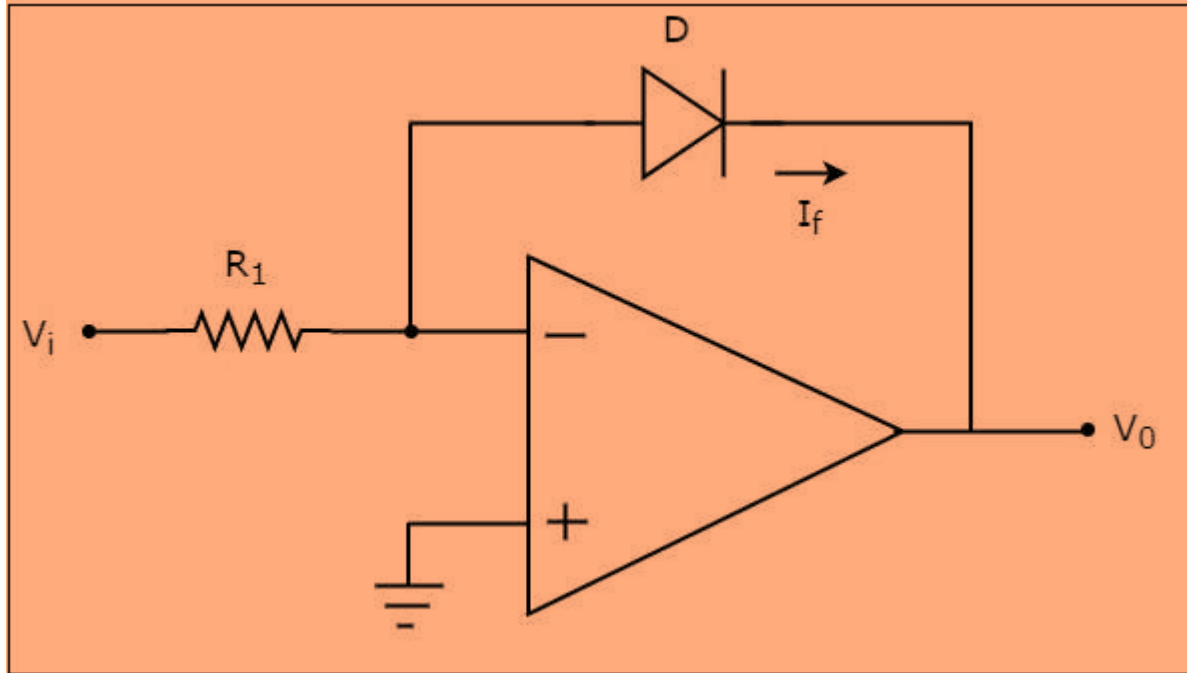


$$V_o = \left(1 + \frac{R_f}{R}\right) \frac{\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Op-Amp: Linear Circuit

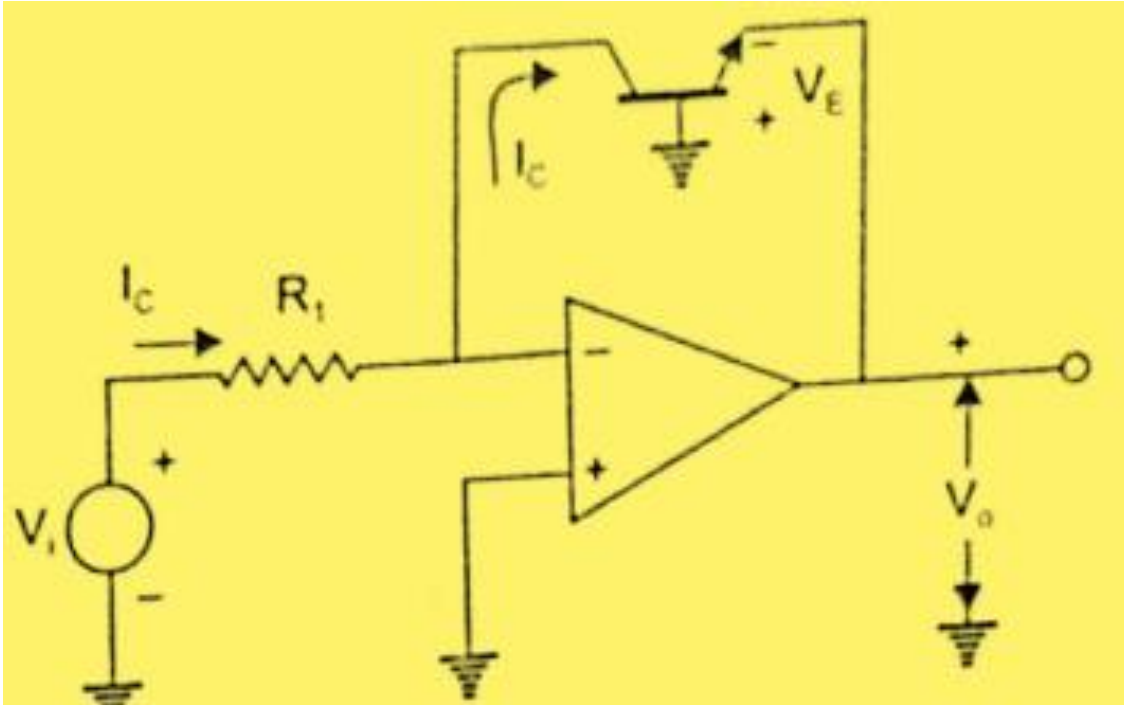


Op-Amp: Linear Circuit

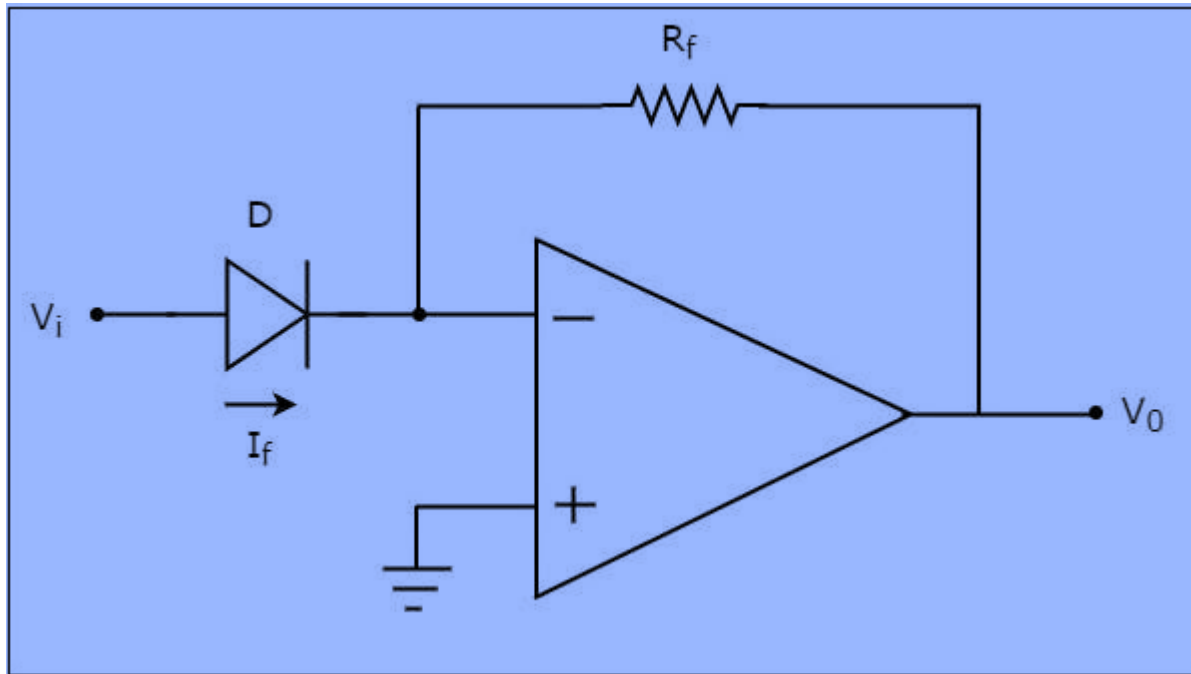


$$V_0 = -nV_T \ln \left(\frac{V_i}{R_1 I_s} \right)$$

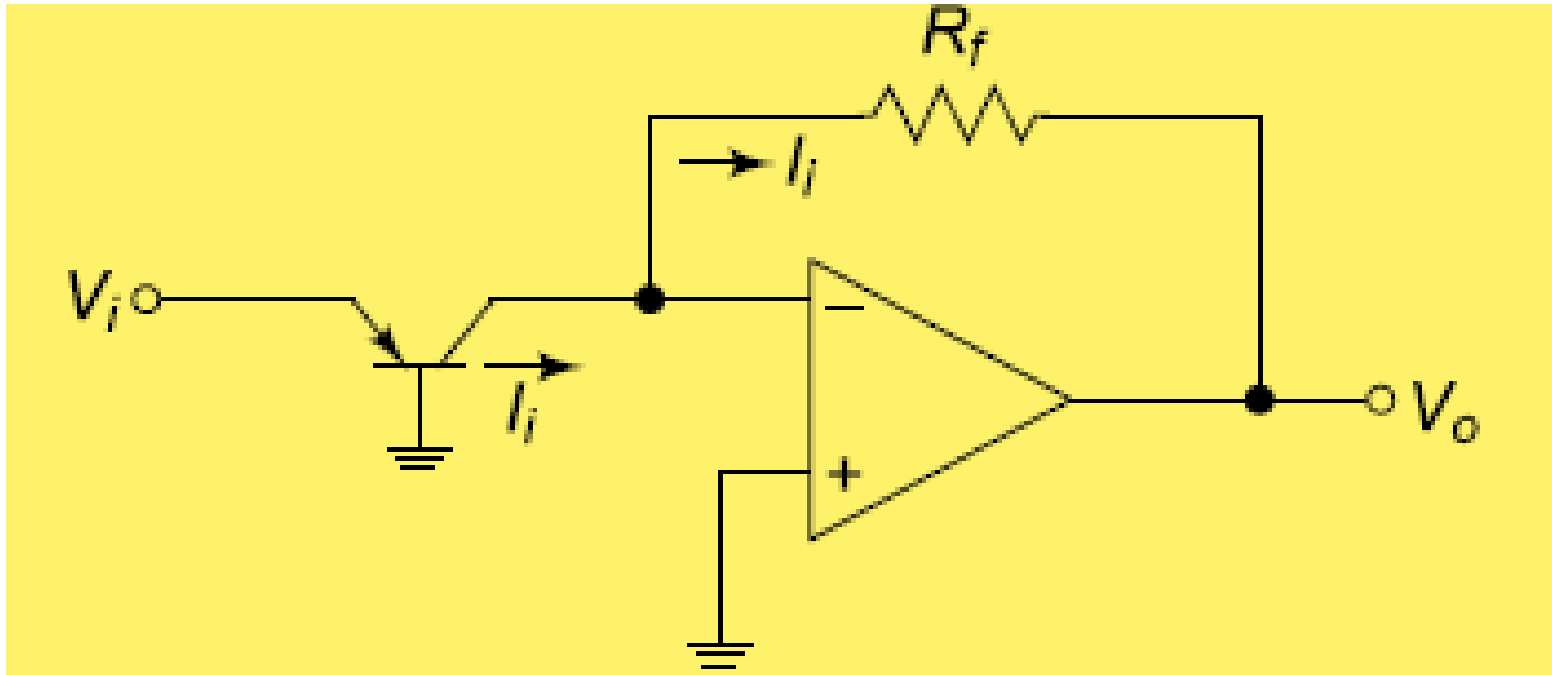
Op-Amp: Log Amplifier



Op-Amp: Antilog Amplifier



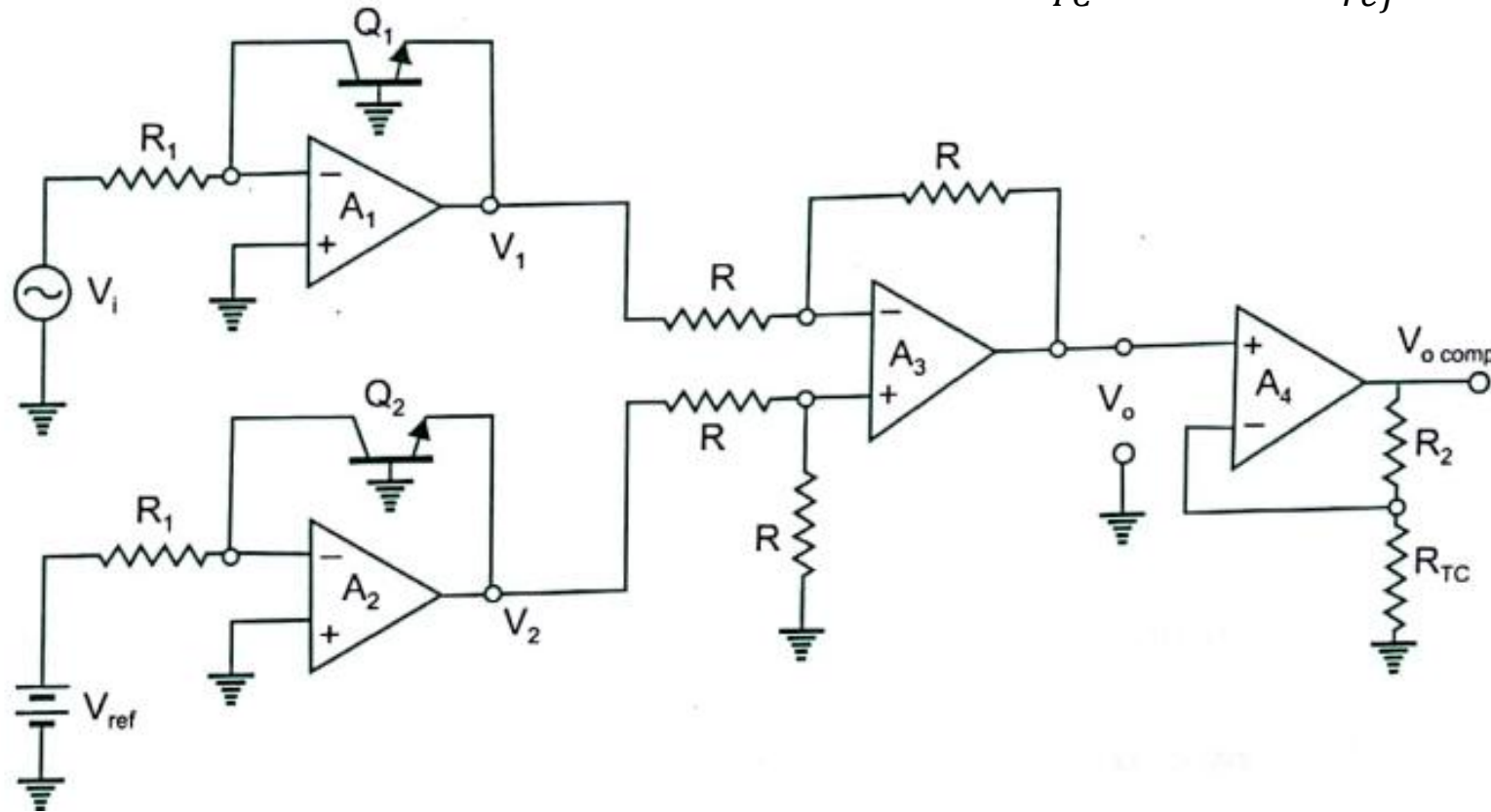
$$V_0 = -R_f I_s e^{\left(\frac{V_i}{nV_T}\right)}$$



$$V_o = -R_f I_S e^{\frac{V_{BE}}{V_T}}$$

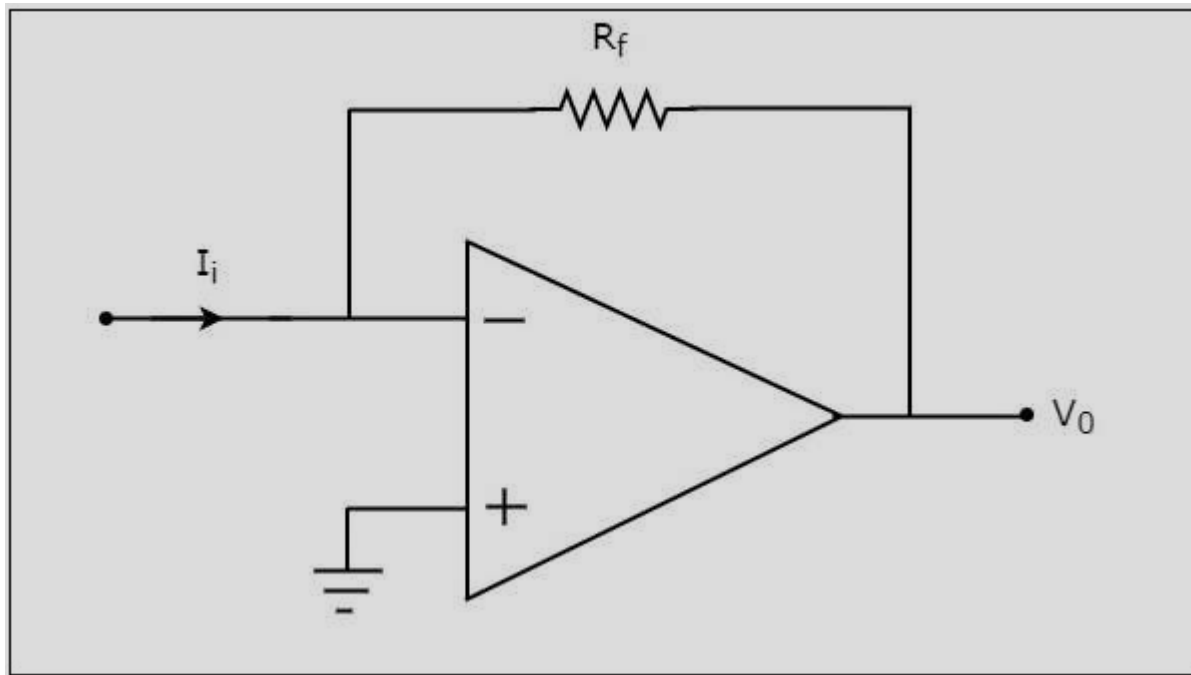
Op-Amp: Compensation

$$V_{\text{comp}} = \left(1 + \frac{R_2}{R_{TC}}\right) \frac{kT}{q} \ln\left(\frac{V_i}{V_{\text{ref}}}\right)$$



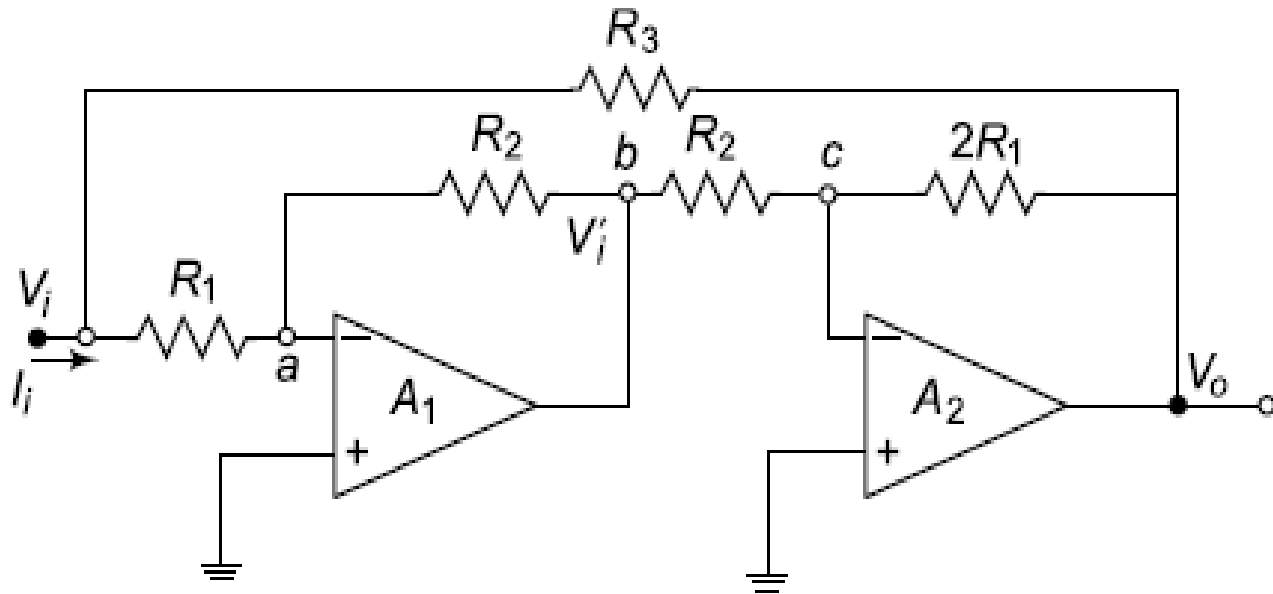
$$V_o = \frac{kT}{q} \ln\left(\frac{V_i}{V_{\text{ref}}}\right)$$

Op-Amp: I-V converter

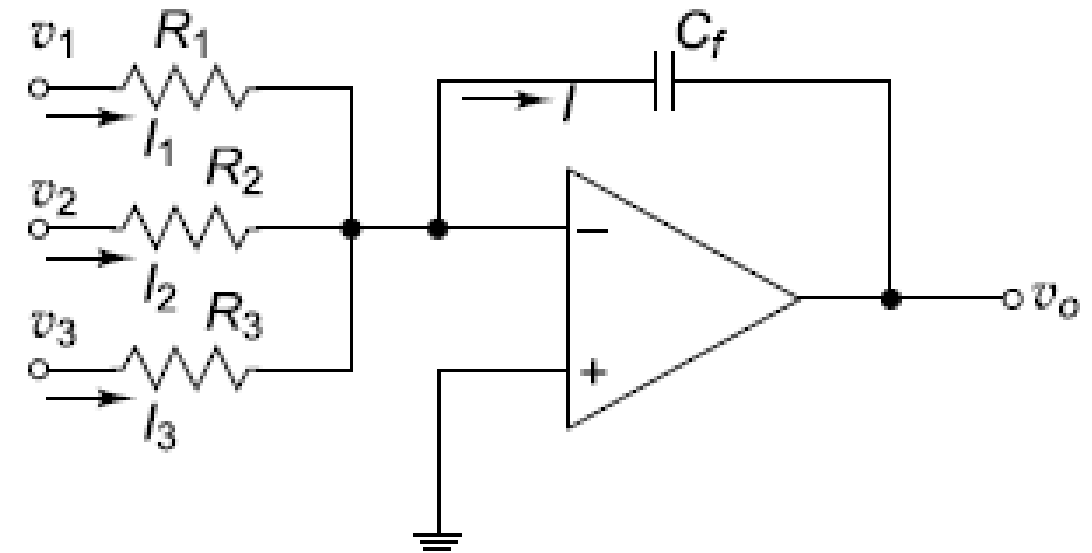


Op-Amp: Example Exercise

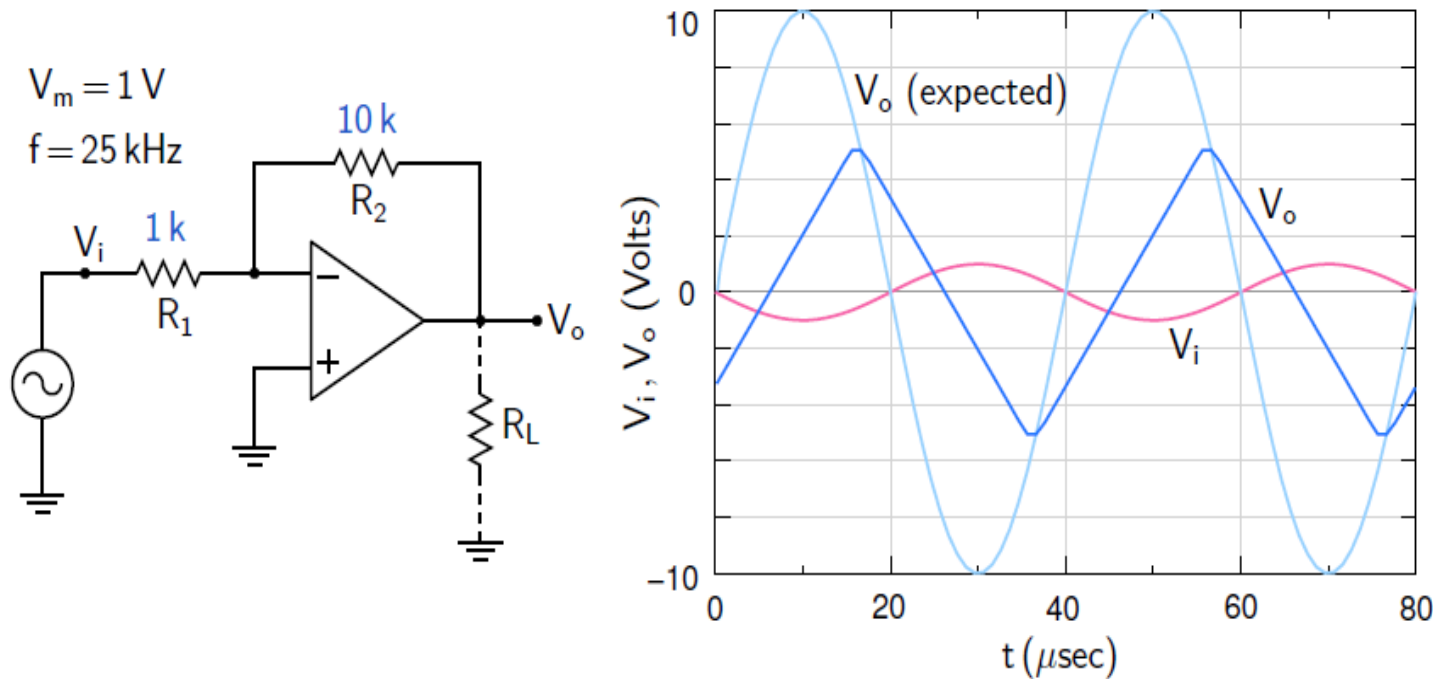
Q.1



Q.2



Op-Amp: Slew Rate

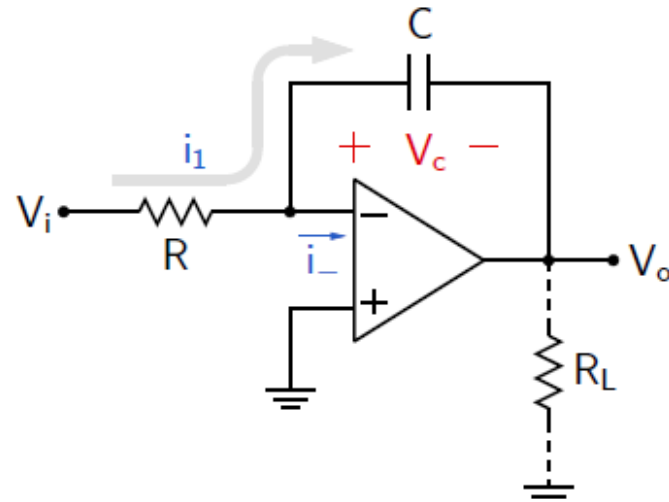


The slew rate (SR) is the maximum rate at which the op-amp can change output without distortion.

$$\text{SR} = \frac{\Delta V_o}{\Delta t} \quad (\text{in V}/\mu\text{s})$$

- * If the signal frequency is too high, a practical op-amp cannot keep up with the input due to its “slew rate” limitation.
- * The slew rate of an op-amp is the maximum rate at which the op-amp output can rise (or fall).
- * For the 741, the slew rate is $0.5\text{ V}/\mu\text{sec}$.

Op-Amp: Linear Circuit



$$V_- \approx V_+ = 0 \text{ V} \rightarrow i_1 = V_i / R.$$

Since $i_- \approx 0$, the current through the capacitor is i_1 .

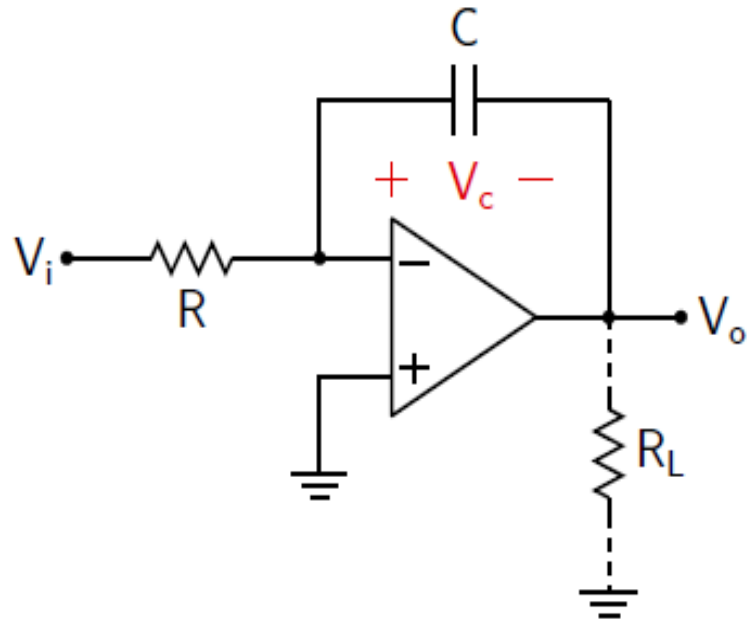
$$\Rightarrow C \frac{dV_c}{dt} = i_1 = \frac{V_i}{R}.$$

$$V_c = V_- - V_o = 0 - V_o = -V_o \rightarrow C \left(-\frac{dV_o}{dt} \right) = \frac{V_i}{R}$$

$$V_o = -\frac{1}{RC} \int V_i dt$$

The circuit works as an integrator.

Op-Amp: Linear Circuit



Given $R=10\text{K}\Omega$; $C=0.2\mu\text{F}$

If $V_o = 0\text{V}$ at $t = 0$, find $V_o(t)$ (Let $t_0 = 0.5\text{ msec}$, $t_1 = 1.5\text{ msec}$).

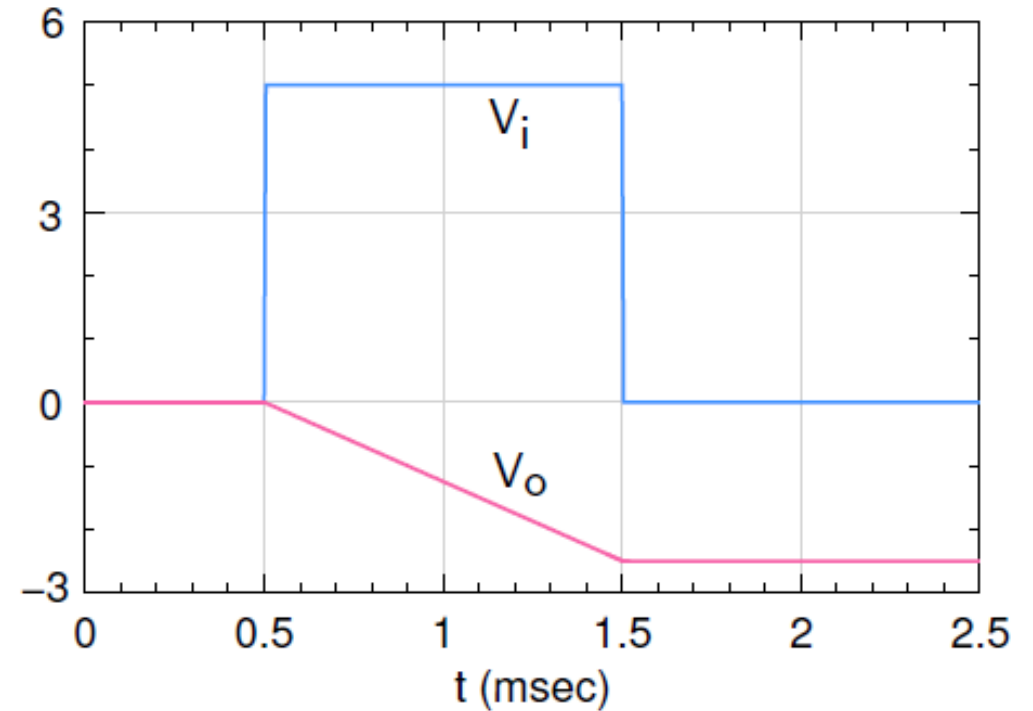
$$V_o = -\frac{1}{RC} \int V_i dt, \quad \tau \equiv RC = 2\text{ msec}.$$

* For $t < t_0$, $V_o(t) - V_o(0) = -\frac{1}{\tau} \int_0^t 0 dt' = 0 \rightarrow V_o(t) = V_o(0) = 0\text{ V}$

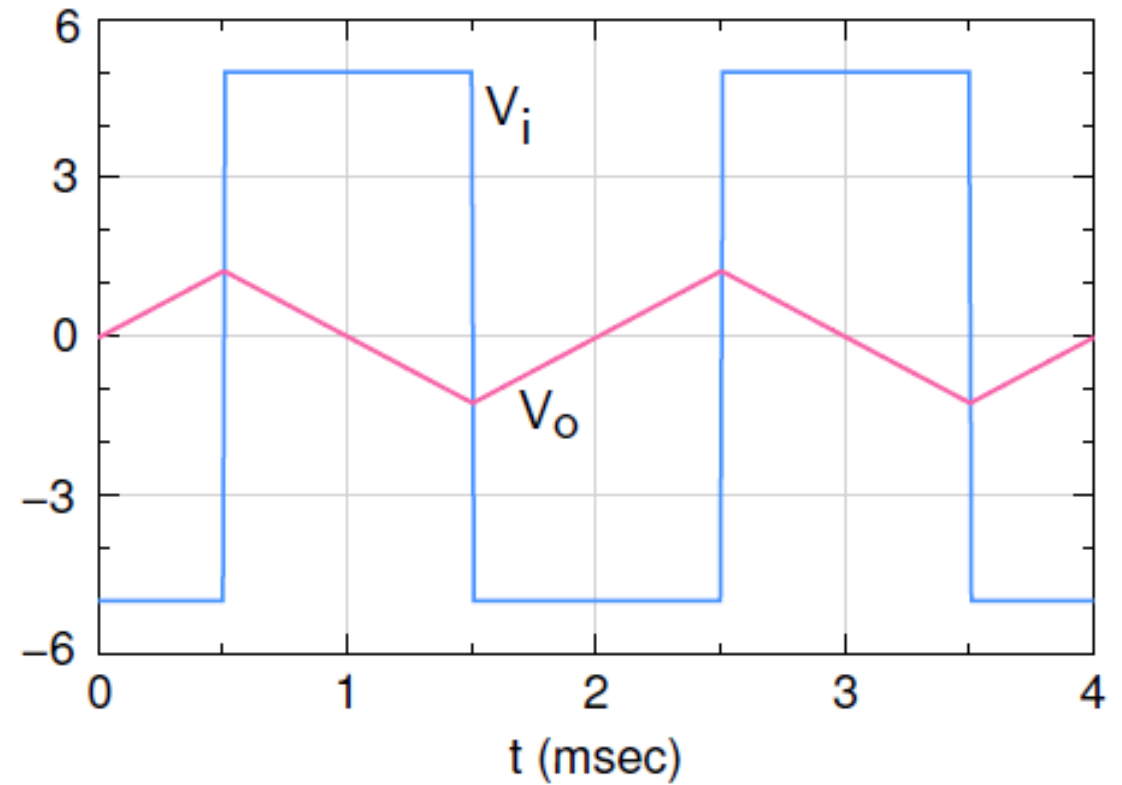
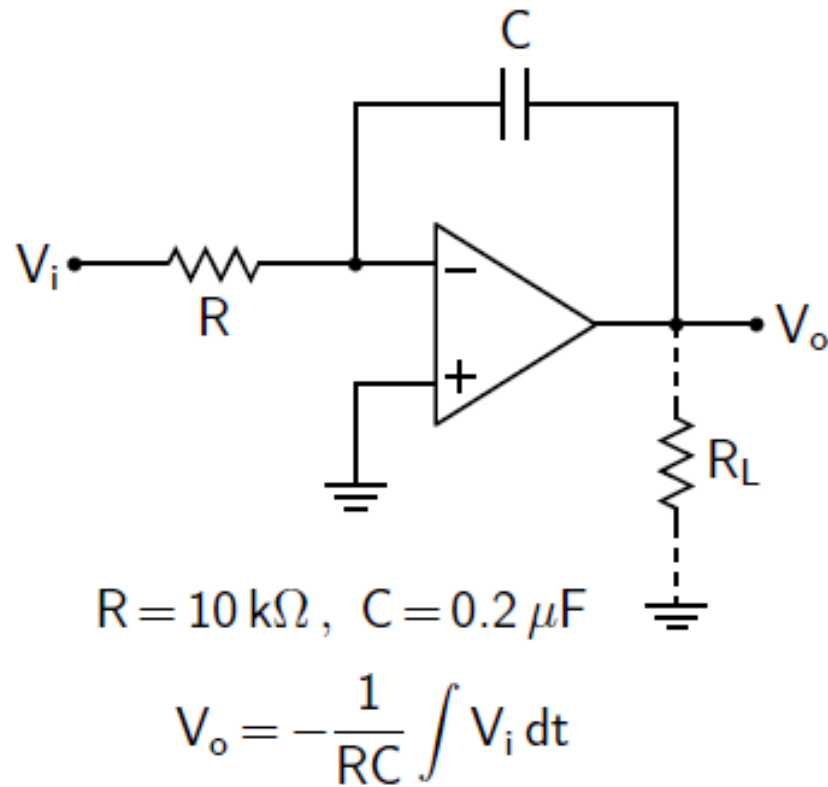
* For $t_0 < t < t_1$, $V_o(t) - V_o(t_0) = -\frac{1}{\tau} \int_{t_0}^t 5 dt' = -\frac{1}{\tau} 5(t - t_0) \rightarrow$ a straight line with a negative slope

At $t = t_1$, $V(t_1) - V(t_0) = -\frac{1}{2\text{ msec}} 5\text{ V} \times 1\text{ msec} = -2.5\text{ V} \rightarrow V_o(t_1) = -2.5\text{ V}.$

* For $t > t_1$, $V_o(t)$ remains constant since $V_i = 0\text{ V}.$

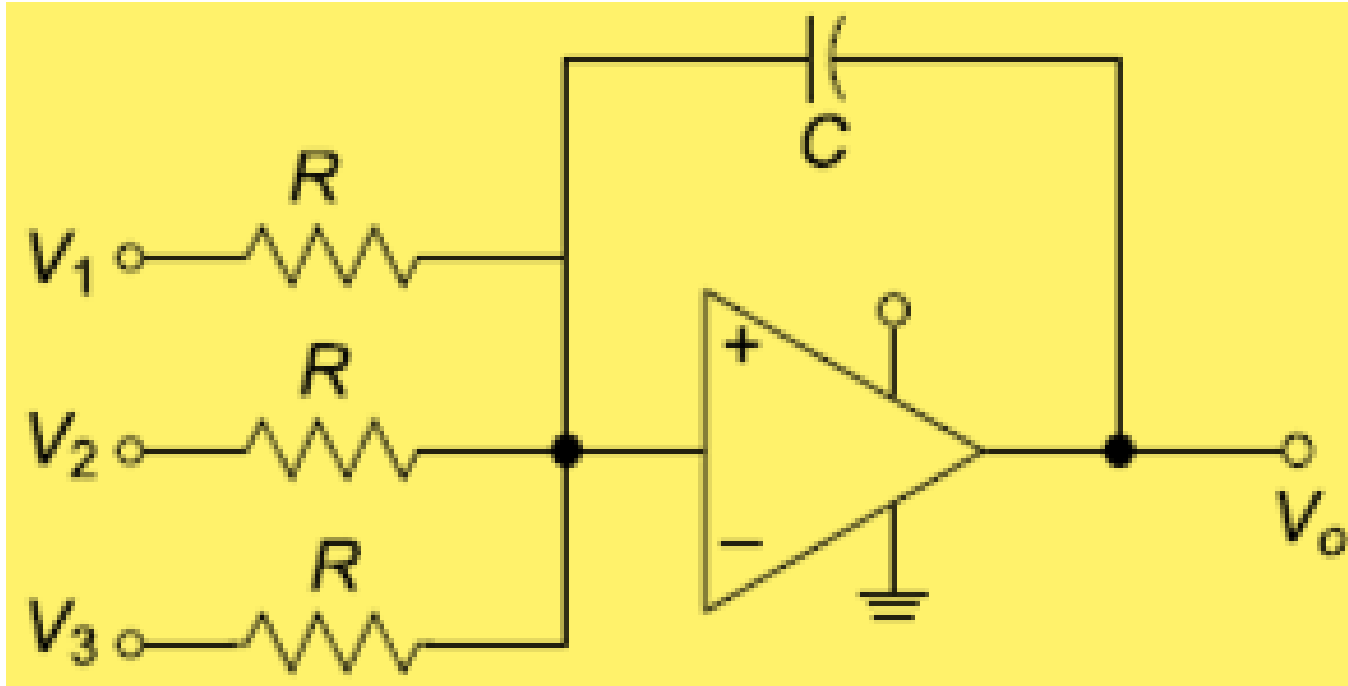


Op-Amp: Linear Circuit

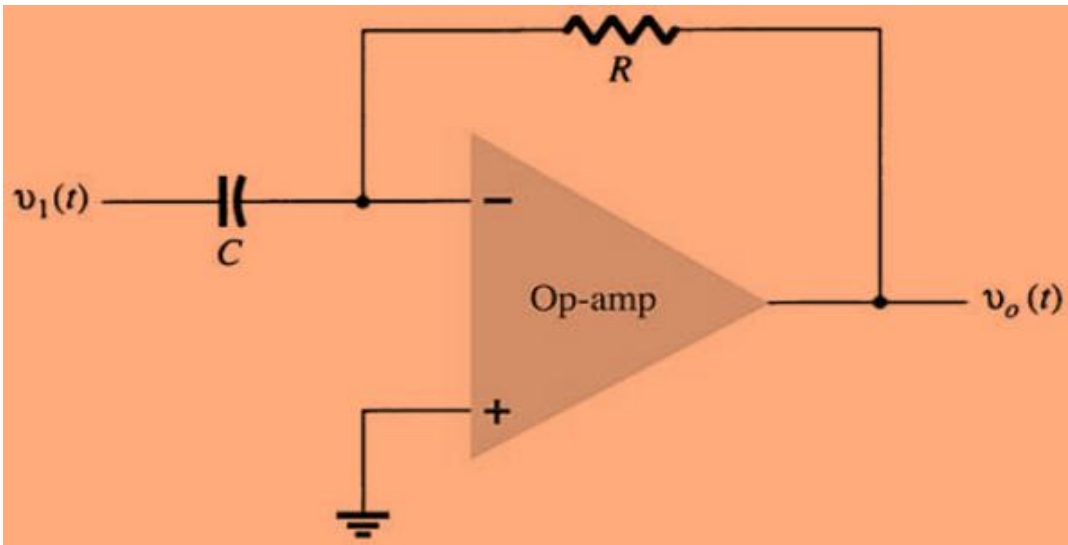


* An integrator can be used to convert a square wave to a triangle wave.

Op-Amp: Summing Integrator

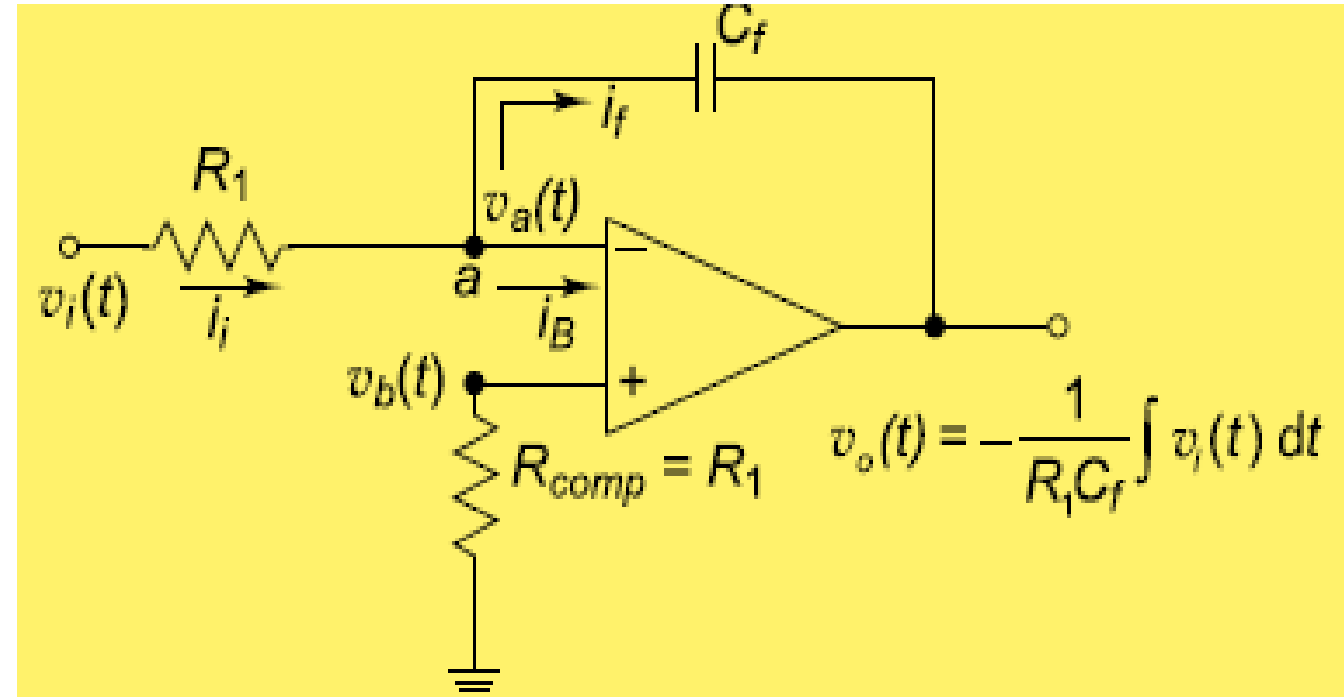
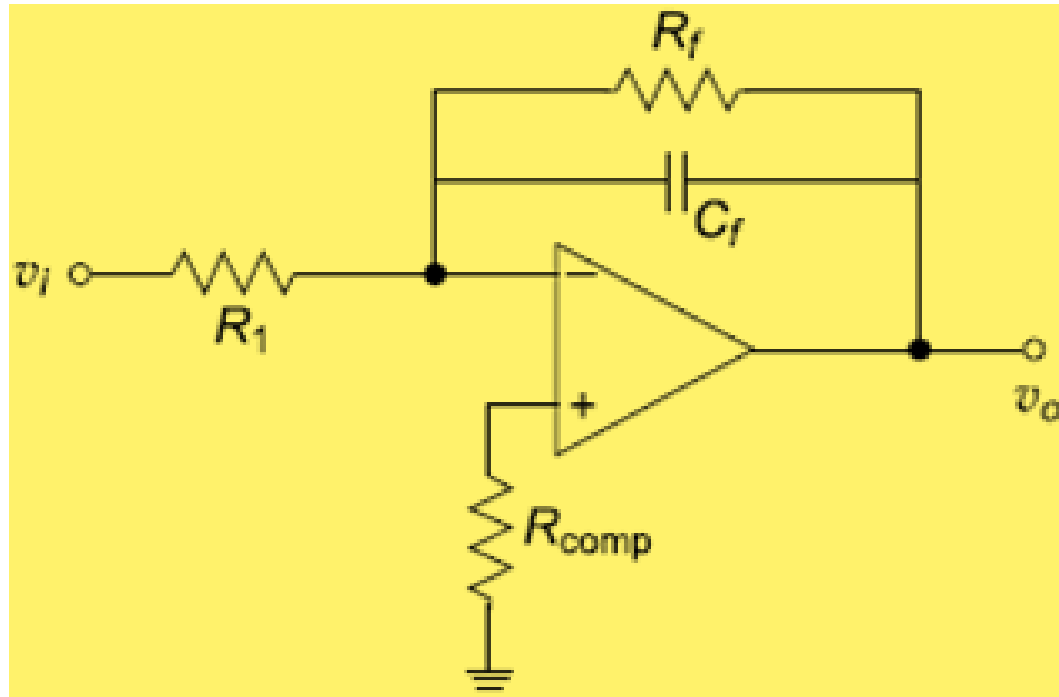


Op-Amp: Differentiator



$$v_o(t) = -RC \frac{dv_1(t)}{dt}$$

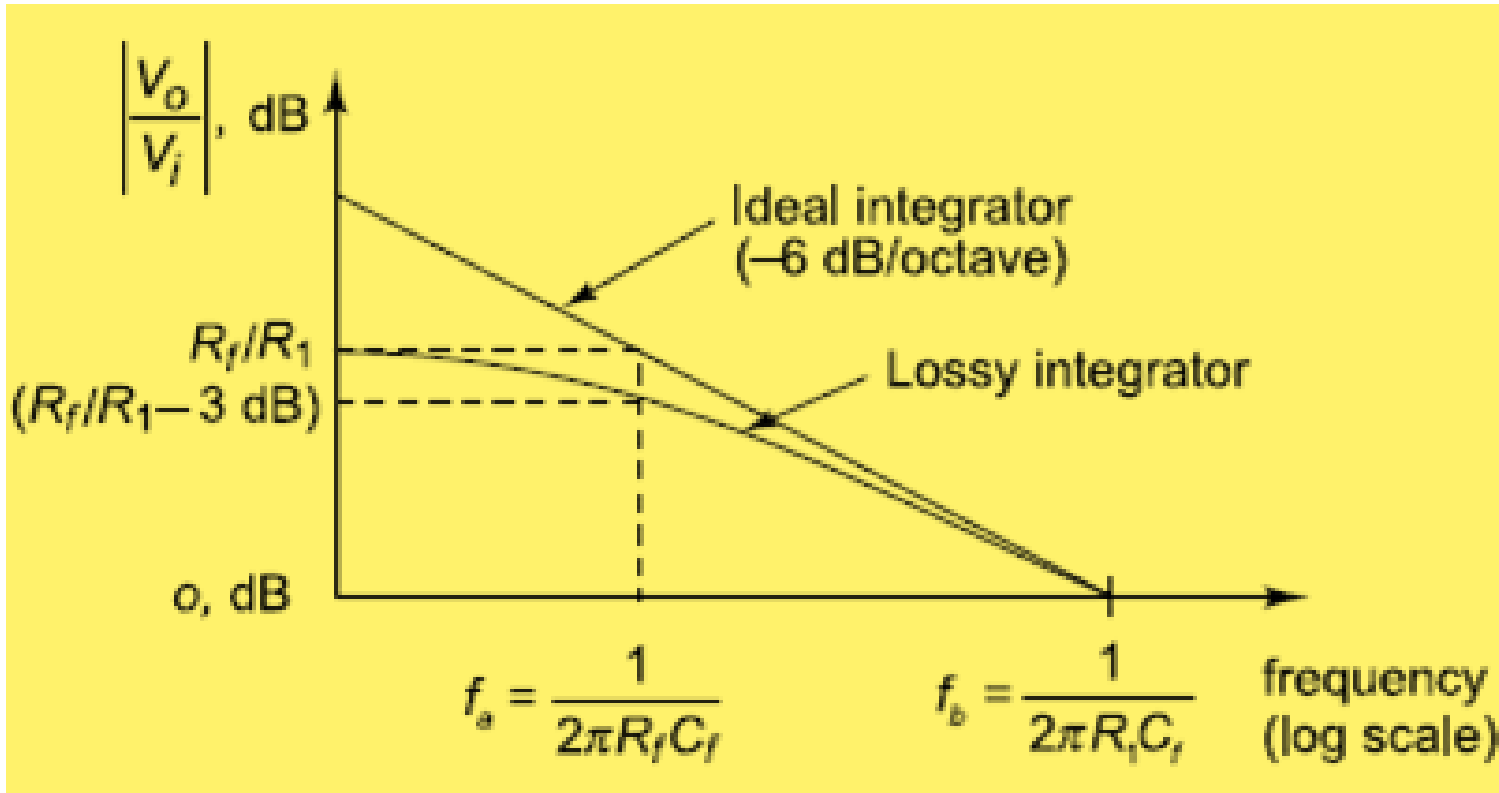
Op-Amp: Practical Integrator vs Ideal Integrator



$$|A| = \left| \frac{V_o}{V_i} \right| = \frac{1}{\sqrt{\omega^2 R_1^2 C_f^2 + R_1^2 / R_f^2}} = \frac{R_f / R_1}{\sqrt{1 + (\omega R_f C_f)^2}}$$

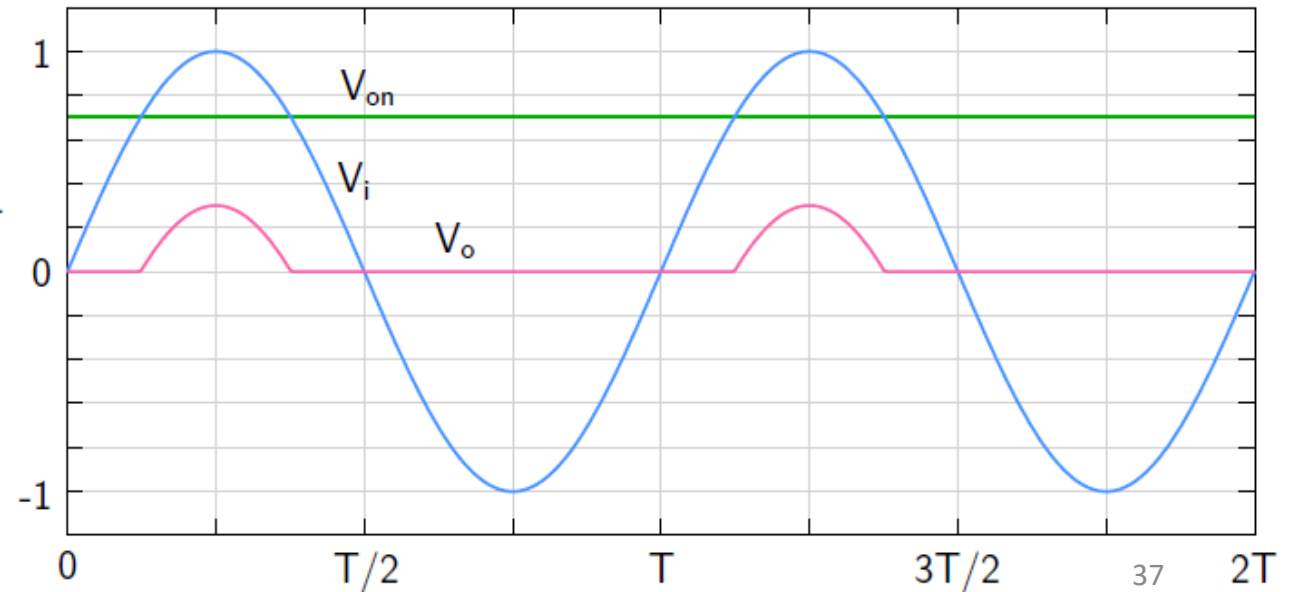
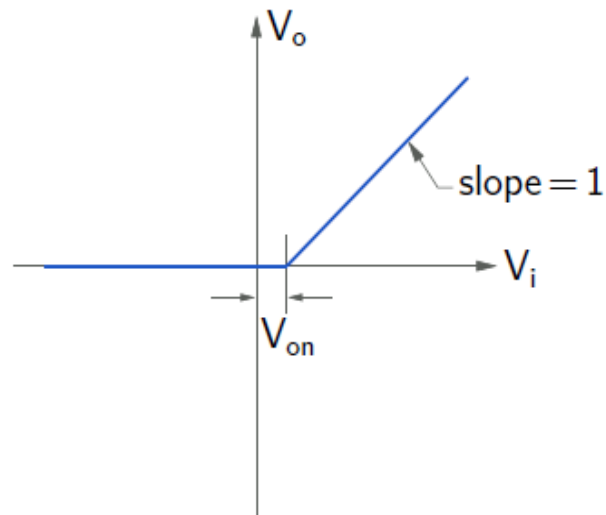
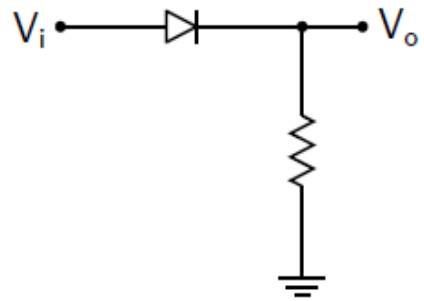
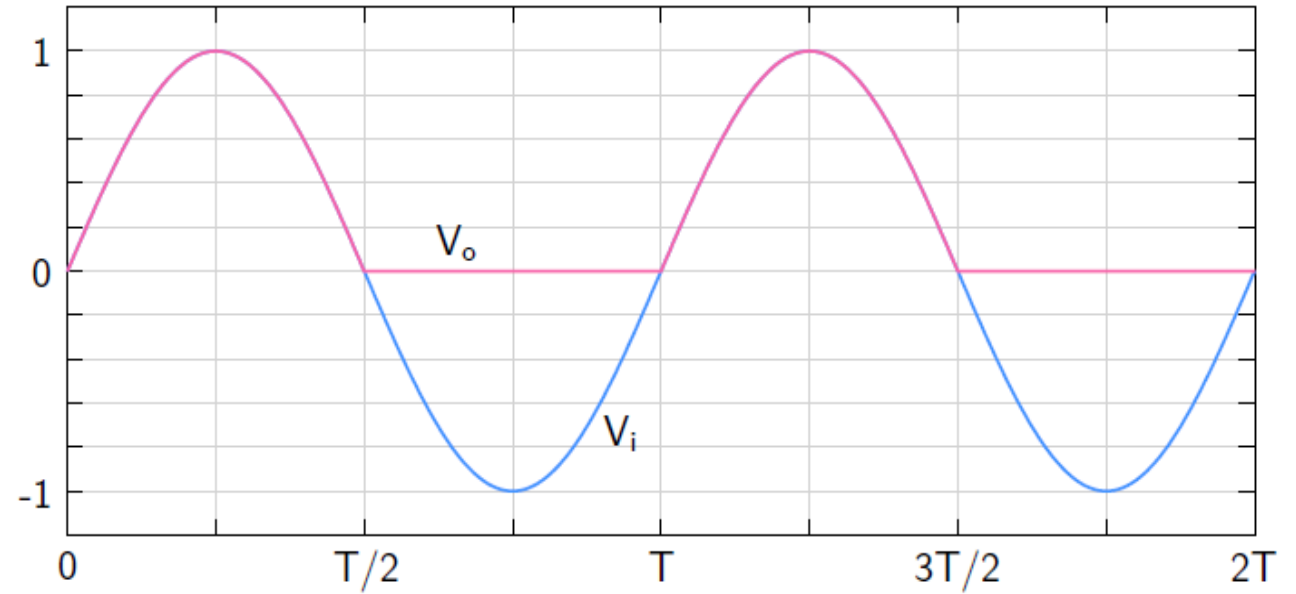
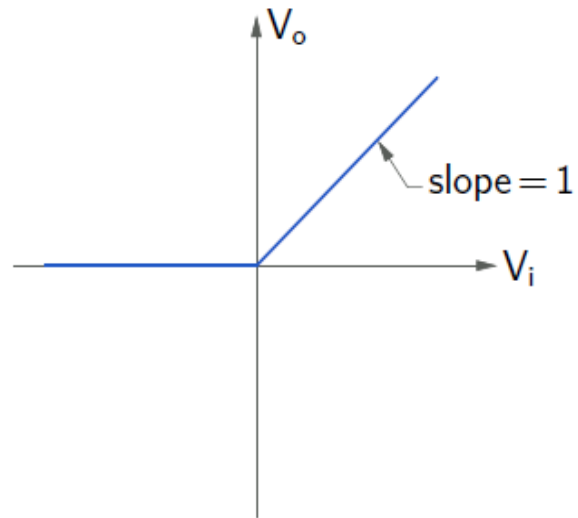
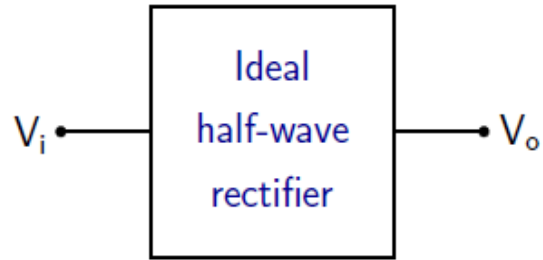
$$V_o(j\omega) = -\frac{1}{j\omega R_1 C_f} V_i(j\omega)$$

Op-Amp: Practical Integrator vs Ideal Integrator



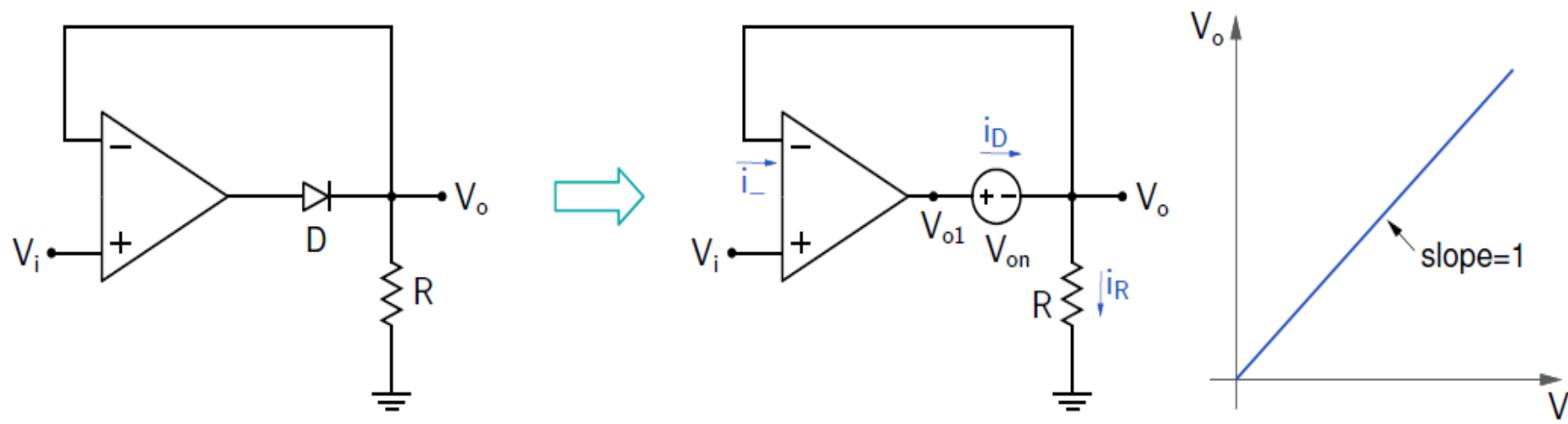
- When the input frequency is less than f_a , the circuit will not act as an ideal integrator and it will only act as a simple inverting amplifier. When the input frequency is equal to f_a , there will be 50 percent accuracy. As a thumb rule, if the input frequency is 10 times f_a , then it results in 99% of accuracy.

Op-Amp: Linear Circuit: Precision Half Wave Rectifier



→ need an improved circuit

Op-Amp: Linear Circuit: Precision Half Wave Rectifier



Consider two cases:

- (i) D is conducting: The feedback loop is closed, and the circuit looks like (except for the diode drop) the buffer we have seen earlier.

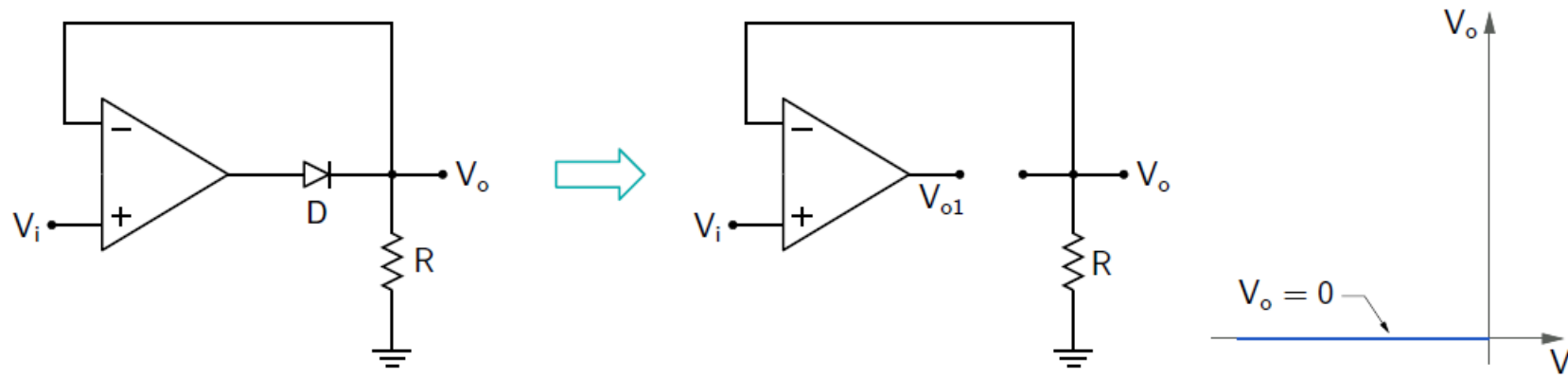
Since the input current $i_- \approx 0$, $i_R = i_D$.

$$V_+ - V_- = \frac{V_{o1}}{A_V} = \frac{V_o + 0.7 V}{A_V} \approx 0 V \rightarrow V_o = V_- \approx V_+ = V_i.$$

This situation arises only if $i_D > 0$ (since the diode can only conduct in the forward direction), i.e., $i_R > 0 \rightarrow V_o = i_R R > 0$, and therefore $V_i = V_o > 0 V$.

Note: V_{on} does not appear in the graph.

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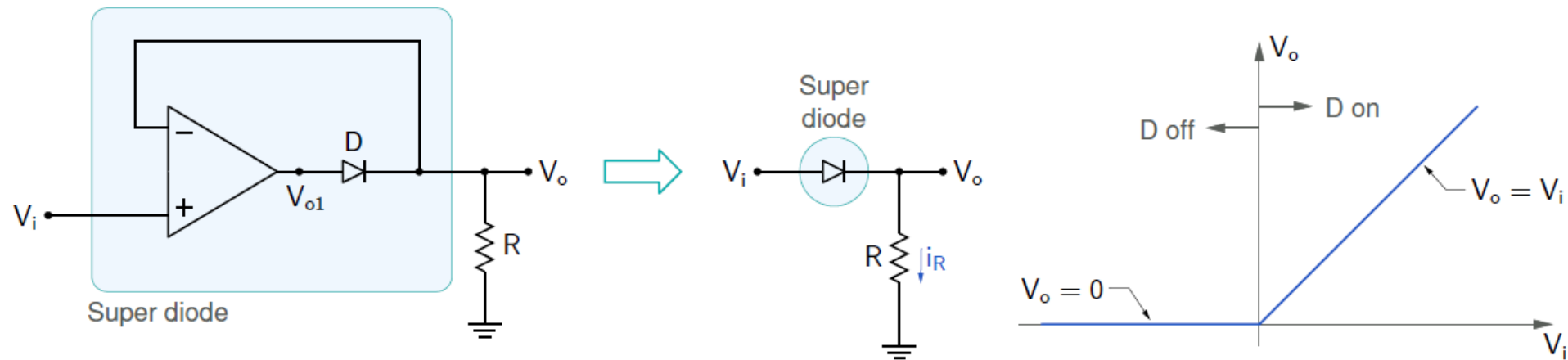
(ii) D is not conducting $\rightarrow V_o = 0\text{ V}$.

What about V_{o1} ?

Since the op-amp is now in the open-loop configuration, a very small V_i is enough to drive it to saturation.

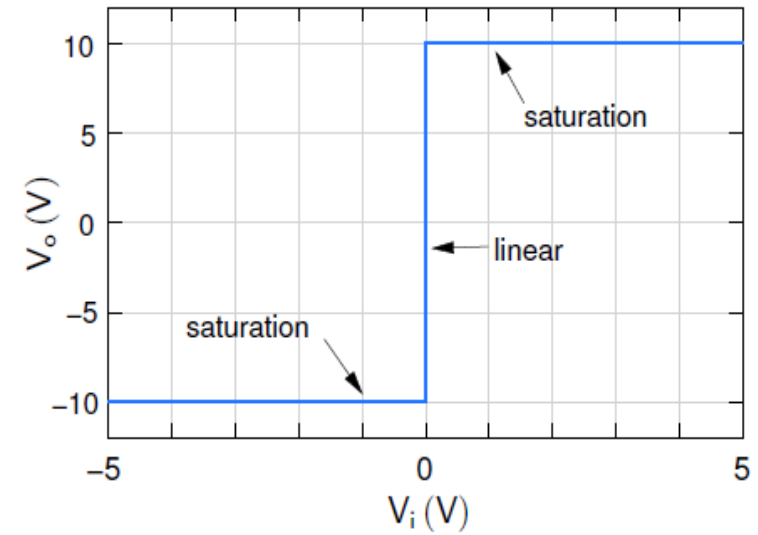
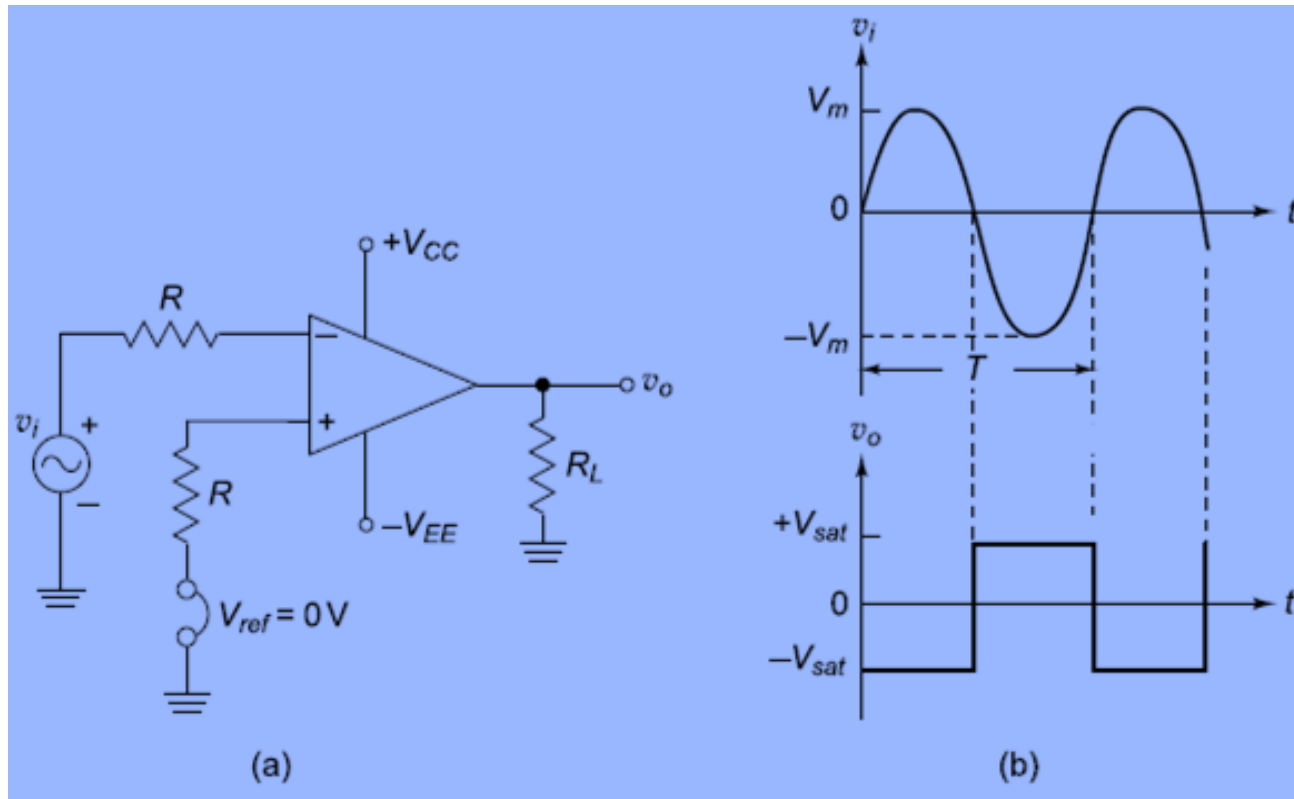
Note that Case (ii) occurs when $V_i < 0\text{ V}$ (we have already looked at $V_i > 0$). Since $V_+ - V_- = V_i - 0 = V_i$ is negative, V_{o1} is driven to $-V_{\text{sat}}$.

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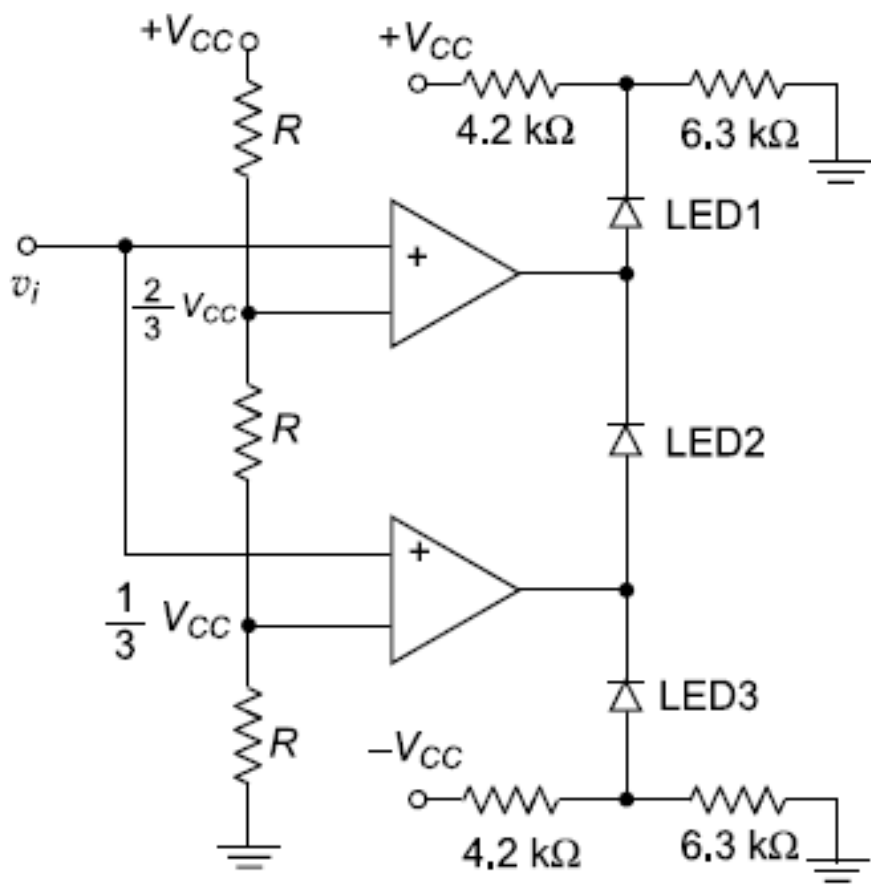


- * The circuit is called "super diode" (an ideal diode with $V_{on} = 0\text{ V}$).
- * When D conducts, the op-amp operates in the linear region, and we have $V_+ \approx V_-$.
- * When D is off, the op-amp operates in the saturation region, $V_- = 0$, $V_+ = V_i$, and $V_{o1} = -V_{sat}$.

Op-Amp: Linear Circuit: Zero Crossing Detector



Op-Amp: Linear Circuit: Window Detector



Three level comparator

Input	LED3	LED2	LED1
Less than 5V	ON	OFF	OFF
Less than 10V and More than 5V	OFF	ON	OFF
More than 10V	OFF	OFF	ON

Op-Amp: Linear Circuit: Filters

- The filters are of two types namely (i) analog filters and (ii) digital filters.
- The analog filters are further classified as passive filters and active filters.
- The passive filters utilize only resistors, inductors and capacitors.
- An active network is a circuit obtained by interconnecting passive elements (resistors and capacitors) and active elements (transistors, tunnel diodes and operational amplifiers).
- An active filter uses an op-amp in order to minimize the effect of loading on the frequency characteristics of the filter.
- The filters are widely used in communication, signal processing and sophisticated electronic instruments.
- The applications of filters also include the suppression of power-line hum, reduction of very low or high-frequency interference and noise, bandwidth limiting and specialised spectral shaping.

Op-Amp: Linear Circuit: Active vs Passive Filters

- (i) The maximum value of the transfer function or gain may be greater than unity.
- (ii) The loading effect is minimal, which means that the output response of the filter is essentially independent of the load driven by the filter.
- (iii) Complex filters can be realized without the use of inductors.
- (iv) The passive filters using R, L and C components are realizable only for radio frequencies. Because, the inductors become very large, bulky and expensive at audio frequencies. The active filters overcome these problems.
- (v) The active filters are easily tunable due to flexibility in gain and frequency adjustments.
- (vi) The op-amp has high input impedance and low output impedance. Hence, the active filters using op-amp do not cause loading effect on the source and load. Therefore, cascading of networks does not need buffer amplifier.
- (vii) Active filters for fixed frequency and variable frequency can be designed easily. The adjustable frequency response is obtained by varying an external voltage signal.

Op-Amp: Linear Circuit: Active vs Passive Filters

The high frequency response is limited by the gain-bandwidth product and slew rate of the practical op-amps, leading to comparatively lower bandwidth than the designed bandwidth.

(ii) The design of active filters becomes costly for high frequencies.

(iii) Active filters require dual polarity dc power supply whereas passive filters do not.

(iv) The active element is prone to the process parameter variations and they are sensitive to ambient conditions like temperature. Hence, the performance of the active filter deviates from the ideal response.